



The Arab American University
Faculty of Information Technology

Project Title: Smart Health Care System

By:

Mohammad Teeti - Major: Computer Science
Hamza Zyoud - Major: Computer Science
Abdallah Badarneh - Major: Computer Science

Supervisor:

Dr. Amjad Rattrout

Date: 15/ 2/ 2025

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1. Front Matter

1.1 Declaration

This document serves to affirm that the project titled "Smart Health Care System" supervised by Dr. Amjad Rattrout, is our original work. And it does not include any unacknowledged contributions or materials previously published or authored by others, except where explicitly quoted in the report.

1.2 Abstract

Modern hospitals often face significant inconveniences in the coordination between the Operation Division, Sterilization Department, Biomedical Equipment Maintenance, and Laboratory functions. Common methods rely on paper-based checklists and isolated information systems, all of this leads to the misplacement of surgical instruments, delays in equipment repairs, and poor management of lab samples. This project presents a solution that utilizes RFID (Radio-Frequency Identification) technology with integrated software modules that work on enhancing the workflows within these departments. The system is designed to track instruments, lab samples, and large medical equipment as they transition between departments. This approach reduces the time needed to find misplaced items, guarantees the integrity and proper handling of samples, the design ensures clear communication across departments without the need to risk time consuming procedures on obtaining necessary information. The Lab module continuously tracks the status of samples and storage temperatures, alerting if conditions break from safe levels. A customized set of dashboards for every role were proposed, to fulfill the personalization features for various user categories, including OR Staff, Sterilization Staff, Maintenance Staff, Lab Staff, Lab Supervisor, and Admin, enhancing both usability and accountability. The architecture is built for growth, enabling the scalability of extra hospital departments to be included if needed. The Smart Health Care System (SHCS) is designed to improve resource efficiency, reduce risk and strengthen patient safety, and deliver a flexible structure that healthcare institutions can modify in response to evolving medical standards and technologies.

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2. Chapter One: Introduction

2.1 Background and Motivation

Hospitals, particularly unlike other facilities, face the pressure of strict safety protocols, tight scheduling, and limited resources. Due to all these limitations, the fast-paced, precise environment of hospitals requires staff to invest time in doing such specific tasks to reduce the error as much as possible, although efforts exist. Recent studies show that up to **6000** hours per month are wasted by hospital staff trying to find misplaced equipment, approximately **16%** of surgeries face the misplaced equipment-related issues.

Operation rooms must operate with flawless precision to avoid these inconveniences, as manual methods for double-checking instruments can lead to significant time waste. Previous statistics demonstrate the costly impact of this issue. All of this can lead to longer patient waiting times, rushed procedures, and an overall strain on the hospital staff.

On the opposite side of the hospital is the laboratory, a section where precision must be impeccable, beginning with the labeling of samples and extending to the preservation of those samples under specific temperature and environmental conditions.

RFID tags can be attached to instruments, large equipment, and samples, automatically updating the system when items transition between hospital departments. By using this technology that can manage workflow and send comprehensive alerts, hospitals can improve patient care and make sure that risk is at an all-time low.

Motivated by these challenges, this project proposes a solution that streamlines pre-operation instrument checks in the OR, eases the procedure of manually verifying the sterilization of instruments upon entry and exit, tracks the status of large equipment, and manages the lab workflow in a convenient way.

2.2 Problem Statement

Even with the continuous advancements in medical technology and workflow optimization systems, many hospitals still lack an effective solution to manage their workflow efficiently.

This raises several worries about the deficiencies in hospital systems, which lie in the presented questions:

- How can hospitals reduce the time staff spend searching for instruments and verifying their availability to minimize last-second searches and patient waiting times?
- How can hospitals enhance sterilization workflows, making sure that instrument sterilization is both timely and thorough, making sure that sterilization is efficient, and ultimately minimizing infection control risks?
- How can hospitals keep OR staff informed about large equipment status, preventing unexpected cancellations and confusion?
- How can lab sample and temperature tracking be monitored to avoid damaged samples or delayed diagnostics?
- How can administrators efficiently manage user roles, guarantee compliance, and uphold reliable backup and recovery procedures?

Justification:

Recent data indicates that thousands of staff hours are lost each month in the search for misplaced equipment, with 16% of surgeries encountering problems directly associated with missing instruments or devices. In the absence of an integrated solution that aligns the workflows of Operating Rooms, Sterilization, Maintenance, and Laboratories, hospitals will face logistical obstacles that affect patient safety, use of resources, and the overall quality of care.

2.3 Aims and Objectives

Aim:

To develop an integrated, RFID-based Smart Health Care System that addresses the core issues in operation division, sterilization, maintenance, and laboratories by providing real-time tracking, automated alerts, simplified communication, and assured administrative oversight.

Objectives:

- Implement Real-Time Tracking & Verification:
 - Set up instruments, trays, and large medical devices with RFID tags to provide immediate location view and automate a checklist verification
- Improve operations and ensure the safety of patients.
 - Minimize the duration staff spend locating and verifying instruments, thus lowering wait times for patients while minimizing the probability of incomplete surgical trays.
- Optimize the sterilization process.
 - Apply automation for the logging of instruments entering and exiting the Sterilization department, categorizing items as “Pending Sterilization,” “In Progress,” or “Ready for Transfer,” while making sure everything is audited for safety and verification purposes.
- Improve Maintenance Coordination
 - Update the status of large equipment (“Under Repair,” “Available”), allow Maintenance staff to record repairs, and notify OR staff of device unavailability for alternative organizing.
- Enhance Lab Sample Management
 - Track samples throughout different laboratory stages ("In Progress", "Priority") while constantly keeping track of critical temperatures for storage and providing immediate alerts for any results that differ from normal ranges.
- Deliver detailed alerts and notifications.
 - Provide real-time notifications for absent instruments, deficient trays, sterilization completion, or irregular temperature readings.
- Support Admin Oversight & Data Integrity
 - Create access control based on roles, maintain full audit logs of all user activities, provide backup and restore functionalities.

2.4 Contributions

This project aims to provide an integrated hospital management solution that gathers the operations of several departments into a singular digital framework:

- Easily combines Operating Room, Sterilization, Maintenance, and Laboratory functions. Every role has a specific dashboard, therefore minimizing unconnected procedures.
- Uses RFID scanning to verify the presence or absence of required instruments, while recognizing any differences between real and necessary checklists for each operation.
- Guarantees that instruments are tracked for status updates (e.g., “Pending Sterilization,” “Ready for Transfer”) and that large equipment repairs are recorded with comments and timestamps, improving consistency and transparency.
- Offers real-time alerts for missing items, sterilization completion, and irregular laboratory conditions ensuring staff can respond quickly to preserve patient safety.
- Tracks all actions, sterilization processes, equipment state updates, and changes to user roles in a secure audit log.
- Creates an adaptable structure capable of integrating additional technologies and easily scalable to other hospital divisions.

3. Chapter Two: Literature Review

3.1 Overview

Hospital operations often face difficulties due to the complex organization of operating rooms, sterilizing processes, maintenance schedules, and laboratory administration. Many software solutions are available to manage certain aspects of these processes, such as operating room scheduling, lab reservation, and the integration of hospital information systems. But these systems don't have an all-in-one way to track instruments in real time, track temperature of samples, and keep an eye on maintenance, especially through an integrated RFID infrastructure. This review will look at existing hospital and healthcare systems from different regions, CompactSoft (Egypt), Efio (Indonesia), NPİSTANBUL (Turkey), MDAD Tech (Egypt), and Medingate, and then compare them to our proposed SHCS.

3.2 Existing Systems Overview

This section offers a brief overview of existing healthcare software solutions available in the market, analyzing their advantages, disadvantages, and overall significance to the goals of the proposed system. By analyzing these established platforms, we can pinpoint effective strategies, steer clear of common challenges, and enhance our own design to meet the unfulfilled requirements in hospital operations.

3.2.1 CompactSoft – Hospital Operations Management System

Description:

CompactSoft offers an organized method for registering and categorizing operating room numbers, operations, and tracking bookings for these rooms. Its main objective is optimizing the scheduling aspect of hospital operations.

Strengths:

The system enables coordinated data management with clear labeling and documentation for rooms and surgeries. It also offers precise tracking of processes which helps in managing booking changes and avoids room conflicts.

Weaknesses:

It focuses mostly on planning and room control instead of providing full real-time instrument tracking or an entire end-to-end operational workflow. In addition, its user interface may appear outdated in comparison to modern, online, or mobile options.

3.2.2 Eflo – Lab Room Booking Application

Description:

CompactSoft is an organized system for recording and sorting operating room numbers, operations, and tracking reservation for these establishments. The main objective is to improve the planning part of hospital operations.

Strengths:

The system allows organized data management through clear labelling and documentation for rooms and operations. It provides precise track of operations, simplifying handling of booking changes and avoiding room conflicts.

Weaknesses:

The system offers a narrow application, limited only to room scheduling and without full lab inventory or equipment tracking features. It could face scalability challenges if the laboratory plans to expand.

3.2.3 NPİSTANBUL – Operating Room Medical Registrar System

Description:

NPİSTANBUL integrates with HIS and Laboratory Information Systems to provide medical staff with quick use of patient data during operations. It aims to gather data for operating room staff, connecting different hospital systems.

Strengths:

The system enables solid data integration by combining OR data with existing hospital or laboratory systems, which provides a complete view of operations. It provides quick access to important patient information, improving medical staff's ability to make choices.

Weaknesses:

Combining with various applications can be challenging and may require considerable instruction or modification. In addition, significant setup efforts is needed to sync the system with the distinct procedures and previous systems of each facility.

3.2.4 Medingate – Laboratory Management System

Description:

MDAD Tech, Medingate focuses on lab operations sample management, test scheduling, and real-time data reporting.

Strengths:

The system includes a user-friendly interface with the goal of improving laboratory processes from sample registration to test ending. It designs easily with full MDAD Tech solutions that allow exchange of information between departments and was developed with a focus on keeping up with regional medical standards and laws.

Weaknesses:

It lacks powerful automated analysis or advanced features like continuous temperature monitoring or RFID-enabled real-time notification.

3.3 Conclusion

Most existing health management systems focus only on isolated tasks, which leads to significant limitations in actual time equipment tracking, sterilization oversight, repair organizing, and laboratory organization. The SHCS combines all of these processes into a single RFID-based system.

3.4 Comprehensive Comparison of Systems

Feature	CompactSoft	Efio	NPISTANBUL	Medingate	SHCS
Primary Focus	OR room scheduling & Labeling	Lab room scheduling	Integration with HIS & Lab data in the OR	Dedicated lab management system	Fully integrated (OR, Lab, Maintenance, Sterilization) via RFID
OR Scheduling	Yes	No	Partial (Focus on data integration)	No	Yes (real-time checks)
Real-Time Instrument Tracking (RFID)	No	No	No	No	Yes
Sterilization Workflow	No	No	No	No	Yes
Maintenance / Equipment Tracking	No	No	No	No	Yes
Lab Sample Management	No	Partial (Room booking only)	Partial (HIS data integration)	yes	Yes (Full sample lifecycle & status)
Real-Time Temperature Monitoring	No	No	No	No	Yes
Key Strength	Structured OR scheduling	Simple, QR-based lab booking	Seamless data integration with HIS	Easy-to-use interface for labs	End-to-end approach across OR, Lab, Maintenance, Sterilization.
Overall Coverage	Focused on OR scheduling	Lab scheduling only	OR data integration	Good for labs	Comprehensive (OR, Lab, Maintenance, Sterilization, RFID)

4. Chapter Three: System Analysis

4.1 Introduction

This section highlights the key aspects of our system and the interactions of stakeholders with it. We will begin with user stories to guide us towards functional requirements that precisely address our needs and encompass the relevant topics we are tackling.

4.2 User Stories

The User Stories part outlines the main user groups and their practical engagements with the proposed system. By understanding users' perspective and defining the objectives and challenges for each position, from Operating Room Staff to Lab Supervisors, we create a focused, user-oriented viewpoint that guides the functional requirements and design choices.

4.2.1 OR Staff Story

A nurse of the OR team logs in to the Smart Health Care System. They must sign up and make an account if they don't already have one. Once they are logged, they navigate to the "Instrument Verification" via their specific dashboard. The OR staff checks the system to see when the next surgery is scheduled so they know what instruments are required via a pre-loaded checklist. When instruments get into the OR it gets scanned via a Reader that scans the room for any RFID tags. The system finds the instruments automatically and matches them to the check list. The staff are alarmed immediately if an instrument is missing. If there is an extra instrument, the staff can decide if they need to take it out or keep it. This can be used also by the staff to keep track of all the instruments that leave the operating room (OR) after surgery and make sure that none are left behind. If any issue occurs with the instrument getting in or leaving the OR, the system will notify the staff via their dashboard. The system keeps track of all actions, like logins, scans, status changes (location of the instruments), missing-instrument alerts, and more, so that every part of the operation is seen, and holding accountability of the OR staff.

4.2.2 Sterilization Staff Story

The sterilization staff have their own interface with the right privileges needed. After logging in, they can see a real-time list of the instruments coming in from the OR into the sterilization department. When a set of used instruments comes in, the system scans them through an RFID reader. The list of the upcoming instruments then is compared with the list of the instruments that left the sterilization room. If any discrepancy occurs (Missing tools, Extra tools), the system alerts the staff about it. After the issue is solved (if occurred) The instruments are then marked as "Pending Sterilization" Then, the staff can supervise the sterilization process and the system automatically changes the instruments state from "Pending Sterilization" to "Sterilization Complete," via RFID readers in a temporary storage, System then notifies OR staff of these ready to use instruments. All changes are logged and recorded for future reference if needed.

4.2.3 Maintenance Staff Story

The maintenance staff use a special interface that shows them a list of all the big medical equipment (like portable imaging devices and specialized surgery tools), with the location of this equipment and its state ("Under Repair", "Available", or any additional notes). Other departments have access to view the state of this equipment; this lets another department know if a key device is suddenly unavailable. Hospital departments such as the OR, Sterilization, and Maintenance have RFID readers installed. If an OR staff asks for the device while it is still being worked on, the system alerts them of its unavailability. Once the repair is complete, the staff finalizes the notes and marks the device as "Available". For an audit record, the system keeps track of these changes.

4.2.4 Laboratory Staff Story

Lab staff access the laboratory staff dashboard. Staff without an account must sign up and log into it. After logging in, they navigate to the Sample Management interface. Staff scan new samples, add test notes, and can link them to a patient ID or a patient file. The system marks the sample as "In Progress" and allocates it to testing sections. One sample can undergo several tests. The system continuously tracks temperatures in essential storage facilities like refrigerators and freezers. Staff are notified about any abnormal change in temperature thresholds. The dashboard also maintains each sample's real-time location, enabling staff to track its testing progress. When testing is finished, the system marks the sample as "Ready" and notifies relevant staff. Staff can point a sample as "Priority", these samples are processed faster. The system logs all staff actions, logins, sample scans, location changes, and test status updates, for full traceability and lab efficiency.

4.2.5 Laboratory Supervisor Story

The Lab Supervisor oversees the whole lab procedures using higher permissions. After login in, they get a real-time dashboard of sample statuses, noted problems (such as incomplete data, possible temperature variations). Should irregularities or errors indicate test results, the supervisor investigates them and decides whether to call for retesting or approval of the findings. The supervisor can also produce comprehensive reports on benchmarks including turned-round times and test results. These investigations increase general laboratory efficiency and help to find repeating issues.

4.2.6 Admin Story

The system grants the highest level of access to administrators. They can schedule backups, apply patches or feature upgrades, control user accounts and responsibilities, create global settings, and control the RFID hardware. For instance, only Lab Supervisors view the Supervisor Dashboard, only OR staff members view the OR Dashboard, etc., therefore ensuring that the role-based access of the system stays accurate. Administrators also handle new equipment or instrument registrations using RFID tags in the system. Their role also includes critical processes like data backup and restore, therefore ensuring that the system maintains compliance and data integrity. All administrative acts are recorded in an audit log.

4.3 User Scenarios & Task Cards

This section shows a series of practical scenarios showing how various users can interact with the system in their daily tasks. By determining specific tasks and pointing out the required steps to achieve them, these scenarios show the system's capabilities in real-world contexts and serve as a reference for both development and testing processes. The proposed scenarios originate from the previous User Stories and have been divided into five sections for enhanced clarity and convenience: Starting Situation, Normal Flow of Events, What Can go Wrong, Other Concurrent Events, and Task Cards for each scenario. This format allows the obtaining of functional requirements that meet the actual needs of the users.

4.3.1 Scenario 1: OR Staff – Login/Registration & Automated Instrument Verification

Starting situation:

Another operation is ready to begin. OR staff members must log into the Smart Health Care System. Under regular events, they either complete the registration form (name, email, password, role) or log in with appropriate credentials. Once logged in, they then check the necessary instruments for the operation.

Normal flow of events:

Once logged in, the staff go to Instrument Verification. If placed inside the range of the RFID reader (inside of the OR), instrument trays are immediately matched to the checklist of the procedure. If instruments existing in the OR room match the required instruments for that specific operation, the system shows a "Ready" mark.

What can go wrong:

If there is a discrepancy between the list and present instruments, for example, missing or excess instruments, the system alerts the OR staff about it to address it.

Other concurrent activities:

OR members can be checking many sets of instruments for simultaneous operations. Or even adding separate instruments if needed, the maintenance staff can update the availability of large equipment for upcoming surgeries.

Task Cards for OR Staff (Scenario 1)

Register & Verify Instruments for Surgery	
Task 1	Create or Access User Account 1. Register: Select “Register” and enter your information (name, email, password, role). 2. Log In: Enter your username and password to access the system.
Task 2	Open Instrument Verification 1. From the OR dashboard, navigate to “Instrument Verification.” 2. Confirm that the specific operation checklist is selected.
Task 3	Perform RFID Scan 1. OR staff get the sterilized instruments from the storage inside the OR room 2. The system automatically matches scanned instruments to the operation’s checklist. 3. If there is a discrepancy, add or remove instruments as required.
Task 4	Complete Verification 1. Confirm that all instruments are present for the operation. 2. The system updates each instrument’s status to “In Use.”

4.3.2 Scenario 2: OR Staff – Tracking Instruments Leaving the OR

Starting situation:

After the operation ends, all used instruments must be transferred from the OR to Sterilization.

Normal flow of events:

Under regular events, OR staff choose "Instrument Outbound" in their own dashboard. Trays and instruments leaving the OR get scanned by an RFID reader that automatically detects every instrument. The system checks the instruments against the pre-operation checklist. After that, the system registers the departure time and changes the status of every instrument to "In Sterilization".

What can go wrong:

A mismatch alarm can occur if a mismatch happens, for example an instrument that is left behind in the OR.

Other concurrent activities:

Sterilization staff could be finishing other batches. Maintenance staff may look at whether any big equipment needs any action.

Task Cards for OR Staff (Scenario 2)

Outbound Instrument Scan (Post-Surgery)	
Task 1	Open Instrument Outbound Module 1. Log in to the system and navigate to OR Staff Dashboard. 2. Select “Instrument Outbound” to begin the outbound process.
Task 2	Gather & Scan Outbound Instruments 1. Collect all used instruments in their trays. 2. Pass the trays and instruments into the sterilization department. 3. The system scans instruments automatically.
Task 3	Occurred Mismatches 1. If the system flags any instruments as missing or extra. 2. Confirm if the item has been left in the OR. 3. Fix the tray contents to match the pre-operation list.
Task 4	Finalize Outbound Process 3. The system records a departure time for each instrument and updates their status to “In Sterilization”.

4.3.3 Scenario 3: Sterilization Staff – Registering Instruments in Sterilization

Starting situation:

The sterilization staff receive an alert indicating that instruments are joining the department from the operating room.

Normal flow of events:

The sterilization staff logs in and examines the Incoming Instruments list. Upon the arrival of instruments, the system verifies the instruments against the list of instruments that left the sterilization department into the OR. Instruments getting in the sterilization department are signed in a “Pending Sterilization” state. Staff then go through the standard sterilization protocol. When sterilization is complete, the system automatically signs the set of instruments as “Ready to Use”, while the system records timestamps for all status modifications, and alerts the operating room upon completion of sterilization.

What can go wrong:

The original list of instruments that left the sterilization department may not match the list of instruments arriving from the operating room. Instruments might not complete the full sterilization process (forgotten in a washing area, for example).

Other concurrent activities:

The OR staff may be assembling new instrument sets for upcoming operations. The maintenance staff may be preparing the equipment requested. The sterilization might be preparing a set of requested instruments.

Task Cards for Sterilization Staff (Scenario 3)

Receive & Sterilize Incoming Instruments	
Task 1	Receive & Register Instruments 1. Log in and navigate to “Incoming Instruments”. 2. The system scans the upcoming instruments and compares it with the instruments that lift the sterilization. 3. Mark instruments as “Pending Sterilization.”.
Task 2	Perform & Complete Sterilization 1. Transfer trays to the sterilization sections. 2. After sterilization is complete, the system marks the instruments and trays as “Ready” and notify the OR about it. 3. Store instruments in the sterilization storage for the OR staff to pick up. 4. The system records timestamps of entry time and status updates times.
Task 3	Occurred Mismatches 1. If the system flags any instruments as missing or extra. 2. Confirm if the item has been left in the OR. 3. Fix the tray contents to match the pre-operation list.

4.3.4 Scenario 4: Maintenance Staff – Tracking Large Equipment**Starting situation:**

An upcoming procedure needs a portable imaging device, but the staff doesn't know if it's being repaired or is ready to use.

Normal flow of events:

The staff in OR checks the "Equipment Tracking" interface to see what the last known state of the device was. If the state is "Under Repair," then they know that the specific device is not ready for use, they might look for an alternative or contact the maintenance staff for further details. When the repairs are done, maintenance staff change the device's state to "Available." RFID readers record the movement of the device to the OR and change its

state to "In OR." The system also records each movement with a time stamp.

What can go wrong:

The device could be found in a place that wasn't expected, which would cause a mismatch alert. For example, somewhere where readers don't exist.

Other concurrent activities:

The OR staff could be checking instruments for a surgery that is happening at the same time, and Maintenance could be working on several pieces of equipment from different areas.

Task Cards for Maintenance Staff (Scenario 4)

Equipment Status & Maintenance Updates	
Task 1	Check & Update Equipment Status 1. Navigate to the Maintenance dashboard and log in. 2. Locate the relevant equipment from the equipment list. 3. Update the status to "Under Repair" or "Available" as needed.
Task 2	Starting Position 1. After receiving equipment into the department edit the state of it to under repair or available as needed 2. If the equipment needs repairs, you can add a note for later.
Task 3	Perform Maintenance 1. Identify the device needing maintenance. 2. Go on with the repairs, cleaning, or servicing as needed. 3. Document the actions in the system: date, time, and notes on completed tasks
Task 4	Finishing Maintenance 1. Change the status of the equipment to "Available" or "Under Repair" as needed. 2. If device is "Available" the department requesting it gets a notification that its ready to use.

4.3.5 Scenario 5: Lab Staff – Login/Registration, Processing, & Assigning Tests

Starting situation:

A group of fresh samples shows up at the lab, ready to be put to the test. The staff need to sign in or create an account if they don't already have one.

Normal flow of events:

The lab staff log in (or first register if they need to). It is possible to scan each sample and link it to the right patient information on the Sample Management page. The sample is marked with the tests needed, and its state is changed to "In Progress" by the system. The lab staff that handled the sample are logged into the system records.

What can go wrong:

When sample details are missing, the system asks the lab staff to confirm his input. If a sample is mislabeled, it can be edited or deleted.

Other concurrent activities:

Supervisors may be keeping an eye on the lab's work in real time, and maintenance staff may be keeping track of which pieces of equipment need repair.

Task Cards for Lab Staff (Scenario 5)

Sample Intake & Test Assignment	
Task 1	Register and Assign Tests to New Sample 1. Log in or Register as needed, 2. Navigate to "Sample Management". 3. Scan the sample and assign it to the patient's record. 4. Select required tests and save. 5. The system changes each sample's status to "In Progress."
Task 2	Update Sample Location & Status 1. The system tracks the sample position and status. 2. When tests are completed, staff can change the sample status to "Ready". 3. Once tests are completed, change the sample status to "Ready." 4. The system records all test steps and completions.

4.3.6 Scenario 6: Lab Staff – Real-Time Tracing & Temperature Monitoring

Starting situation:

Samples are kept in temperature-controlled storage, like freezers or refrigerators, and the lab system must keep track of their temperature.

Normal flow of events:

Sensors send temperature data to the lab staff dashboard in real time. If the temperature deviates from the normal thresholds, the system alerts the lab staff. They act on the alert (for example, by checking the storage unit).

What can go wrong:

If a sample is stored in the wrong location based on its assigned tests, the system will notify the staff about the location mismatch (e.g., if it is placed in the wrong freezers or refrigerators).

Other concurrent activities:

Supervisors may be looking over temperature logs or looking into repeated deviations, while lab workers are working on other samples normally at the same time.

Task Cards for Lab Staff (Scenario 6)

Monitor Samples Conditions	
Task 1	Set Up Temperature Thresholds 1. The Lab supervisor Logs in and navigate to “Temperature Settings” 2. Sets up acceptable ranges based on test types or storage conditions. 3. Save changes to apply new thresholds.
Task 2	Track Sample Placement 1. The system updates real-time location for each sample. 2. If a mismatch occurs address, it by moving or relabeling.
Task 3	Monitor Temperature Alerts 1. If sensor readings go out of range, the system automatically alerts the staff. 2. Lab supervisor Investigate the cause, and act accordingly. 3. Record the corrective action in the system.

4.3.7 Scenario 7: Lab Supervisor – Oversight, Issue Resolution, & Reporting

Starting situation:

The supervisor keeps an eye on everything going on in the lab from a single dashboard that shows the state of samples, issues that have been flagged by the staff, and important alerts, and have the ability of changing thresholds.

Normal flow of events:

The supervisor logs in with more rights than usual and view the dashboard, where they can see "In Progress," "Ready," or flagged samples, as well as temperature alerts. They then investigate any flagged records (for example, missing sample data or repeated temperature deviations). The supervisor can also generate reports on throughput, turnaround times, or samples data as needed. All these actions are recorded.

What can go wrong:

A lab staff can flag a sample with some notes pointing out that this sample is unusable, then supervisor does the required procedures to solve that issue.

Other concurrent activities:

The lab staff is still working on the samples, and Admin may be in charge of setting up the system or making backups.

Task Cards for Lab Supervisor (Scenario 7)

Supervise Lab Operations	
Task 1	Review Temperature Logs 1. Check historical temperature data to review issues. 2. Generate a report for auditing. 3. Solve flagged issues if they exist.
Task 2	Resolve Flagged Issues 1. Select “Flagged Issues” from the supervisor dashboard. 2. Investigate each flagged sample. 3. The system shows the staff who inputs a flag issue for reference with its notes.
Task 3	Adjust Thresholds 1. Navigate to “Temperature Settings” to modify parameters. 2. Confirm changes and apply them globally.

4.3.8 Scenario 8: Admin/IT – System Configuration, Backup & Recovery, and User Management

Starting situation:

Administrators oversee the whole system, including setting up RFID tags, managing user permissions, backing up data, and configure hardware readers.

Normal flow of events:

Admin logs in to the Admin Dashboard and manages user accounts by approving new staff registration and removing access from former staff. They also set up RFID readers and tags and give each instrument, large equipment, or sample a unique ID. Admin sets up or starts backups of data. An audit log keeps track of all the important changes.

What can go wrong:

A tag on a specific instrument stop working, admin works on it immediately by removing the old tag and configuring a new one.

Other concurrent activities:

The OR staff, Sterilization, Maintenance, and the Lab continue with their daily jobs, and Supervisors may ask Admin to fix problems that affect the lab system.

Task Cards for Admin (Scenario 8)

Admin System Oversight & Configurations	
Task 1	Configure System Settings 1. Log in as Admin and open “System Settings.” 2. Adjust OR/Lab checklists, RFID reader configurations, or sensor thresholds. 3. Save changes; the system logs the update.
Task 2	Manage Users & Roles 1. Select “User Management” from the Admin Dashboard. 2. Add new users, edit roles, or remove access as needed. 3. The system records these actions in the audit log.
Task 3	Backup & Recovery 1. Go to “Backup/Restore” and confirm or schedule new backups. 2. Optionally test a restore in a secure environment. 3. Ensure the system logs backup/restore actions for compliance.
Task 3	Equipment & Instrument Registry 1. Navigate to “Asset Management” and choose “Add New Item.” 2. Input instruments details (name, type, RFID tag ID). 3. Save then system logs all of the updates.

4.4 Requirements Gathering

This section outlines the techniques and procedures employed to pinpoint and enhance the system's requirements, utilizing user narratives, practical situations, and input from stakeholders. By methodically gathering the requirements of every user group, we guarantee that the solution is both pertinent and thorough.

4.5.1 Functional Requirements

This subsection details the specific functionalities the system must deliver, informed by the insights obtained during the requirements gathering process, to fulfill both user and organizational needs. Every requirement pertains to a specific feature or behavior, establishing the groundwork for the following design and implementation processes.

4.5.1.1 User Requirement

1. OR Staff Requirements

1. Login & Registration

- (UR-OR-1.1) Staff can create an account with details (name, email, password, role).
- (UR-OR-1.2) Staff can log into their account to access the OR dashboard.

2. Instrument Verification

- (UR-OR-2.1) Staff can view required instrument for the selected operation.
- (UR-OR-2.2) Staff can view available instrument and their location and state.
- (UR-OR-2.3) Staff can Add or Remove separated instruments from the Instruments list.
- (UR-OR-2.4) The system scans operation room instruments, detecting and highlighting any mismatches with the required list.
- (UR-OR-2.5) The system displays a 'Ready' sign if all required instruments are present.

3. Outbound Instrument Tracking

- (UR-OR-3.1) Staff can view the system as it detects instruments leaving the OR, ensuring none are left behind.
- (UR-OR-3.2) System logs the departure time of the instruments.

4. Large Equipment Tracking

- (UR-OR-4.1) Staff can view required large equipment for the selected operation.
- (UR-OR-4.2) Staff can view available large equipment and their location and state.
- (UR-OR-4.3) Staff can request or reserve needed devices for scheduled procedures.

5. Notification & Alerting

- (UR-OR-5.1) Alerts appear when trays are incomplete or requested items are missing.
- (UR-OR-5.2) Alerts also appear if a large device is missed or under maintenance.
- (UR-OR-5.3) Alerts also appeared when a sterilization is completed

2. Sterilization Staff Requirements

1. Login & Registration

- (UR-ST-1.1) Staff can create an account with details (name, email, password, role).
- (UR-ST-1.2) Staff can log into their account to access the Sterilization dashboard.

2. Manage Sterilization Workflow

- (UR-ST-2.1) The system marks instruments as 'Pending Sterilization' when they enter the sterilization room from the OR.
- (UR-ST-2.2) The system marks instruments as 'Ready for transfer' when they done the sterilization.

- (UR-ST-2.3) Every status change is automatically timestamped and logged.

3. Receive Instruments

- (UR-ST-3.1) The system automatically verifies that all instruments that left the Sterilization department are returned.
- (UR-ST-3.2) The system highlights any mismatches in instrument counts, allowing staff to review and resolve discrepancies.

4. Batch Processing

- (UR-ST-4.1) Staff can batch-process multiple instrument sets at once.
- (UR-ST-4.2) Each item in the batch retains individual logs for traceability.

5. Notifications & Alerts

- (UR-ST-5.1) Staff receive alerts for incoming items.

3.Maintenance Staff Requirements

1. Login & Registration

- (UR-MA-1.1) Staff can create an account with details (name, email, password, role).
- (UR-MA-1.2) Staff can log into their account to access the Maintenance dashboard.

2. Equipment Maintenance Status

- (UR-MA-2.1) The system asks the staff to sign the equipment as 'Under Repair' or “Available” automatically upon entering the maintenance department.
- (UR-MA-2.2) Maintenance staff can add notes to equipment.
- (UR-MA-2.3) Staff can update a device’s status (Under Repair, Available).

3. Maintenance Scheduling

- (UR-MA-3.1) Staff can see the equipment that needs repair.
- (UR-MA-3.2) Staff can view when equipment entered the maintenance department.

4. Location Tracking

- (UR-MA-4.1) RFID or other sensors detect if equipment has been moved to the maintenance area, or OR.
- (UR-MA-4.2) Maintenance staff can confirm or correct location errors.

5. Reports & Logs

- (UR-MA-5.1) Staff can generate maintenance history reports.
- (UR-MA-5.2) The system logs the entry and exit times of each equipment for compliance audits.

4.Lab Staff Requirements

1. Login & Registration

- (UR-LAB-1.1) Lab staff can create an account with details (name, email, password, role).
- (UR-LAB-1.2) Lab staff can log into their account to access the Lab dashboard.

2. Sample Management

- (UR-LAB-2.1) Staff can register new samples, linking each with patient details and required tests.
- (UR-LAB-2.2) The system updates a sample's status to "In Progress" before it tests begin.
- (UR-LAB-2.3) Staff can mark a sample as "Ready" when testing is completed.

3. Sample Tracking

- (UR-LAB-3.1) The system automatically updates and displays each sample's location.
- (UR-LAB-3.2) Staff can track the real-time status of any sample from the dashboard.

4. Temperature Monitoring

- (UR-LAB-4.1) The system displays current temperature readings for critical storage areas (e.g., refrigerators, freezers).
- (UR-LAB-4.2) Staff receive alerts if readings deviate from acceptable ranges.

5. Priority Mode

- (UR-LAB-5.1) Staff can label samples as "Priority" for expedited processing.
- (UR-LAB-5.2) The system highlights these priority samples on the dashboard.

6. Notifications & Alerts

- (UR-LAB-6.1) The system provides immediate notifications for critical events (e.g., temperature deviations, not found sample).
- (UR-LAB-6.2) Lab staff can escalate serious issues to the Lab Supervisor as needed.

5.Lab Supervisor Requirements

1. Login & Registration

- (UR-LS-1.1) Supervisors can create an account with details (name, email, password, role).
- (UR-LS-1.2) Supervisors can log in to access the Supervisor dashboard.

2. Workflow Oversight

- (UR-LS-2.1) Supervisors can view a consolidated dashboard of all sample statuses, locations, and flagged issues.
- (UR-LS-2.2) Supervisors can review temperature data and investigate deviations.

3. Issue Resolution

- (UR-LS-3.1) Supervisors can investigate and resolve flagged or incomplete data.
- (UR-LS-3.2) The system logs all actions taken for accountability and auditing.

4. Reporting

- (UR-LS-4.1) Supervisors can generate throughput, turnaround time, and usage reports.
- (UR-LS-4.2) Reports can be exported or printed for planning and compliance.

5. Lab Management

- (UR-LS-5.1) Supervisors must approve new Lab staff registrations and confirm assigned roles.
- (UR-LS-5.2) Supervisor configures acceptable temperature ranges and alert thresholds.

6.Admin Requirements

1. Login & Access Management

- (UR-AD-1.1) Admin staff can log in to the System Dashboard using their email and password.
- (UR-AD-1.2) Admin has access to global system settings covering both Lab and OR divisions.

2. User & Role Management

- (UR-AD-2.1) Admin can create, modify, or remove user accounts for all divisions (Lab Staff, Lab Supervisor, OR Staff, Sterilization, Maintenance, IT Staff).
- (UR-AD-2.2) Admin defines role-based access permissions.
- (UR-AD-2.3) All changes to user accounts and roles are logged for audit and security tracking.
- (UR-AD-2.4) Admins approve registrations and grant access as needed.

3. System Configuration

- (UR-AD-3.1) Admin configures RFID readers and tags, assigning them to samples, equipment, and instruments.
- (UR-AD-3.2) Admins configure system thresholds and alerts, including temperature deviations, RFID tag lifespan, and tray sterilization notifications
- (UR-AD-3.3) Admin manages operation-specific checklists by defining the required tools and instruments OR.
- (UR-AD-3.4) Admin can add or remove equipment while assigning corresponding RFID tags.

4. Backup & Restore

- (UR-AD-4.1) The system supports scheduled and manual backups for lab data (samples, test logs, inventory) and OR data (instrument movements, sterilization records, equipment tracking).
- (UR-AD-4.2) Admin can restore data to ensure compliance and operational continuity.

5. Audit & Security

- (UR-AD-5.1) Admins can view a detailed audit trail of sample and instrument movements, temperature readings, and user actions.
- (UR-AD-5.2) All system changes are timestamped for regulatory compliance.

6. Maintenance & Updates

- (UR-AD-6.1) Admin applies system patches, bug fixes, and feature updates.
- (UR-AD-6.2) Admin can manage hardware integrations, such as adding new RFID readers or sensors.

7. Alerts & Notifications

- (UR-AD-7.1) Admins monitor and respond to critical alerts, including RFID tag expirations, sample expirations, and sterilization completions.

4.5.1.2 System Requirement

1. Authentication & Authorization

1. User Account Creation

- (SR-1.1) The system shall allow new users to register with specific details (name, email, password, role).
- (SR-1.2) Admins approve registrations and grant access as needed.

2. Role-Based Login

- (SR-1.3) The system shall provide a login interface requiring valid credentials (email and password).
- (SR-1.4) The system shall restrict dashboard access based on user role, directing users to the appropriate dashboard (Lab Dashboard, Supervisor Dashboard, OR Dashboard, Sterilization Dashboard, Maintenance Dashboard, Admin Dashboard).

3. User & Role Management

- (SR-1.5) Admin shall be able to create, edit, and delete user accounts for both Lab and OR users.
- (SR-1.6) Admin shall be able to assign or revoke roles, with all changes logged in an audit trail.

- (SR-1.7) Supervisor shall be able to create, edit, and delete user accounts for Lab users.

2. OR Operations System Requirements

1. Instrument Requirements & Verification

- (SR-2.1) The system shall store operation-specific instrument checklists called operation type.
- (SR-2.2) Upon selection of an operation, the system shall display the list of required instruments.
- (SR-2.3) OR staff can add separate instruments and trays to the active list for the specific operation.
- (SR-2.4) The system shall display the current state (Available, Under Sterilization, Under Maintenance) and location (OR, Sterilization, Maintenance) of each instrument.

2. Instrument Detection

- (SR-2.5) The system's RFID or scanning mechanism shall detect which instruments are present in the OR.
- (SR-2.6) The system shall compare detected instruments with the required list:
 - If all instruments match, the system sets a "Ready" status.
 - If any required item is missing or an extra item is detected, the system shall highlight the discrepancy.

3. Instrument Check-In/Check-Out

- (SR-2.7) The system shall automatically log instruments leaving the OR with a timestamp.
- (SR-2.8) The system shall automatically log instruments arriving from Sterilization to ensure completeness.

4. Large Equipment Requests

- (SR-2.9) The system shall store a list of large equipment required for specific operations.

- (SR-2.10) OR staff shall be able to request large equipment for scheduled procedures.
- (SR-2.11) The system shall display real-time availability and location of large equipment.

3. Sterilization System Requirements

1. Receiving Instruments

- (SR-3.1) When instruments arrive from the OR, the system shall automatically mark them as “Pending Sterilization.”
- (SR-3.2) The system shall verify the list of incoming instruments, comparing them against the items that left Sterilization previously.

2. Sterilization Workflow

- (SR-3.3) System shall update the instruments’ status to “Ready for Transfer” once they are in the Ready for Transfer area.
- (SR-3.4) The system shall automatically timestamp each status change and log it in a Sterilization log.

3. Mismatched or Missing Items

- (SR-3.5) The system shall notify sterilization staff if any instruments are missing upon arrival.
- (SR-3.6) The system shall automatically store any discrepancies for later investigation.

4. Notifications & Alerts

- (SR-3.7) The system shall display alerts on the Sterilization dashboard for incoming instruments.
- (SR-3.8) The system shall notify OR staff when sterilization is complete, including a list of instruments processed.

4. Maintenance System Requirements

1. Equipment Maintenance Status

- (SR-4.1) Maintenance staff determine the status of equipment upon arrival, marking it as 'Under Repair' or 'Available' in the system.
- (SR-4.2) Maintenance staff shall be able to update equipment status to:
 - Under Repair
 - Available (once it is ready to be used again)
 - Reserved

2. Maintenance Scheduling & Logging

- (SR-4.3) The system shall display a list of equipment currently under repair, along with the date/time they arrived.
- (SR-4.4) Maintenance staff shall be able to add notes to equipment.
- (SR-4.5) The system shall log changes in equipment status and store them for compliance audits.

3. Location Tracking

- (SR-4.6) The system shall detect movement of equipment between OR and Maintenance using RFID.

4. Maintenance Reports

- (SR-4.7) Maintenance staff shall be able to generate or export maintenance history for any piece of equipment, including date of repair.

5. Admin System Functions

1. System Configuration

- (SR-6.1) Admin shall be able to configure RFID tags and readers in the system, linking each tag to a specific instrument or equipment item.
- (SR-6.2) Admin shall manage operation-specific instrument checklists (add, remove, or modify required instruments).

- (SR-6.3) Admin shall configure Lab test catalogs, device integrations (RFID, temperature sensors), and Priority rules.
- (SR-6.4) Admin shall define acceptable temperature ranges for storage units.

2. Backup & Restore

- (SR-6.5) The system shall support automated scheduled backups and manual backups.
- (SR-6.6) Admin shall be able to restore system data from the backup.

3. Audit Logs & Reviews

- (SR-6.7) The system shall record all changes in an audit log.
- (SR-6.8) Admin shall be able to query and filter the audit log based on date ranges, user actions, or instrument/equipment ID.

6. Lab Staff System Requirements

1. Sample Management & Tracking

- (SR-8.1) The system shall allow Lab staff to register new samples with patient data and required tests.
- (SR-8.2) The system shall mark samples as “In Progress” when testing begins.
- (SR-8.3) The system shall track each sample’s location (e.g., storage area, test station) in real time, using RFID.
- (SR-8.4) The system shall display any sample’s latest location on the Lab dashboard.
- (SR-8.5) Lab staff shall be able to mark a sample’s status as “Ready” upon completing all required tests.
- (SR-8.6) The system shall timestamp all status changes for traceability.
- (SR-8.7) The system shall allow staff to mark samples as “Priority” to expedite processing.
- (SR-8.8) Priority samples shall be highlighted distinctly on the Lab dashboard.

2. Temperature Monitoring

- (SR-8.9) The system shall integrate with temperature sensors (refrigerators, freezers) to receive real-time readings.
- (SR-8.10) The system shall store temperature logs at configured intervals.
- (SR-8.11) The system shall trigger an alert if a storage unit's temperature goes outside the acceptable range.
- (SR-8.12) The system shall display these alerts on the Lab dashboard and notify relevant staff or supervisor.
- (SR-8.13) The system shall retain historical temperature data for auditing and compliance reporting.
- (SR-8.14) The system shall allow Supervisor and Admin to query temperature logs by date range or storage location.

7. Lab Supervisor System Requirements

1. Supervisor Oversight & Reporting

- (SR-9.1) Supervisors shall see a consolidated view of sample statuses, temperature logs
- (SR-9.2) Supervisors shall be able to review flagged issues directly from the dashboard.
- (SR-9.3) The system shall allow Supervisors to generate reports on sample throughput, turnaround times.
- (SR-9.4) The system shall allow these reports to be exported (PDF, Excel) or printed.
- (SR-9.5) The system shall allow Supervisors to close or resolve flagged issues, recording comments.
- (SR-9.6) All resolution actions shall be timestamped and saved for audit.

4.3.2 Non-Functional Requirements

1. Product Requirements

1.1 Usability

- (NFR-1.1.1) The interface must be simple and logical, to be easy to use for the user.
- (NFR-1.1.2) The alerts must be conspicuously exhibited so users can respond in 5 seconds.
- (NFR-1.1.3) Foreach functionality should require minimal steps to do it.
- (NFR-1.1.4) Foreach errors messages should contain a clear instruction for the right steps.

1.2 Performance & Responsiveness

- (NFR-1.2.1) The system should respond to user actions within 2 seconds during normal events.
- (NFR-1.2.2) The system should handle ongoing processes without any noticeable lags or inconsistent data.
- (NFR-1.2.3) The system should update and display the real time of instruments, equipment, or lab samples within 3 seconds at least.
- (NFR-1.2.4) The system should display the critical alerts within 5 seconds at least.

1.3 Reliability & Availability

- (NFR-1.3.1) The system should be available 99.9% times of overall hospital hours.
- (NFR-1.3.2) The system should be able to recover from critical failures within 1 hour to avoid significant disturbances.
- (NFR-1.3.3) The system should follow role-based access control and encrypt the data.
- (NFR-1.3.4) The system should record any failures related to sensors or networks or RFID reader, and send an alert to the admin.

1.4 Scalability & Extensibility

- (NFR-1.4.1) The architecture should grant smooth integration with the hospital architecture.
- (NFR-1.4.2) The system should be able to scale and adding new departments or more users without and drop in performance.

2. Organizational Requirements

2.1 System Compatibility & Infrastructure

- (NFR-2.1.1) The system should work consistently within the hospital infrastructure.
- (NFR-2.1.2) The system should grant smooth within the existing hospital system.

2.2 Hardware Durability & Environmental Resilience

- (NFR-2.2.1) RFID readers and temperatures sensors should work under the hospital situations and high humidity.
- (NFR-2.2.2) RFID tags should withstand the hospital sterilization (135°C) and humidity.

3. External Requirements

3.1 Regulatory & Legal Compliance Requirements

- (NFR-3.1.1) The system should comply with the associated data-protections to make sure the patient privacy.
- (NFR-3.1.2) The system should comply with the associated HIPAA for the lab sample and patient privacy.

3.2 Ethical Requirements

- (NFR-3.2.1) The system should ensure processing patient and staff securely and confidentially following the ethical data handling principles.
- (NFR-3.2.2) The system should log all the staff actions to be auditable.

4.6 Use Case Modeling

4.6.1 Use Case Diagram

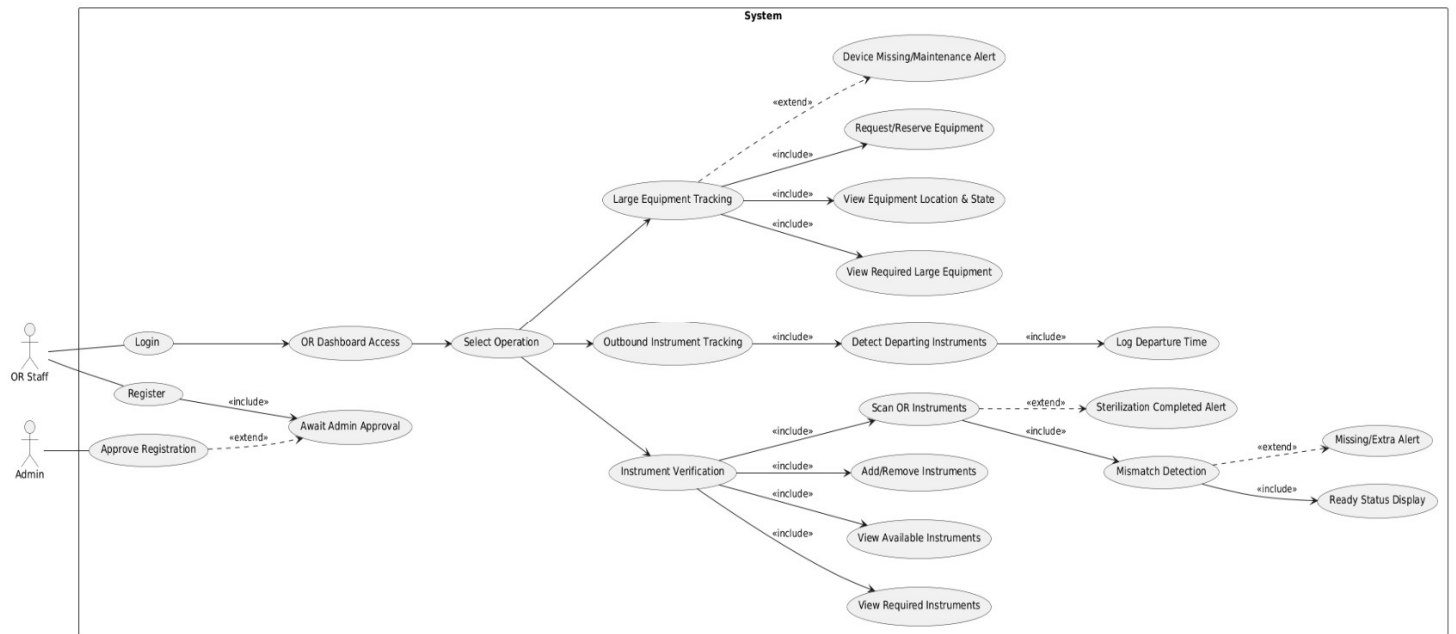


Figure 4.6.1.1: OR Staff Use Case Diagram

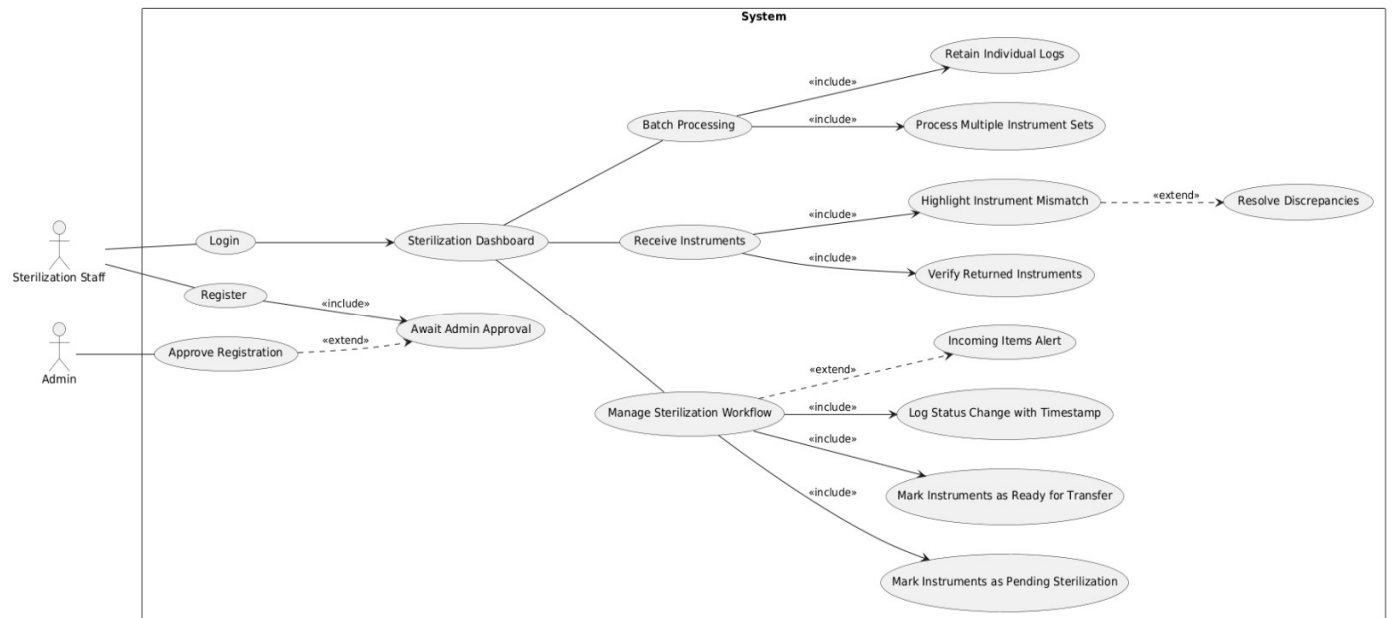


Figure 4.6.1.2: Sterilization Staff Use Case Diagram

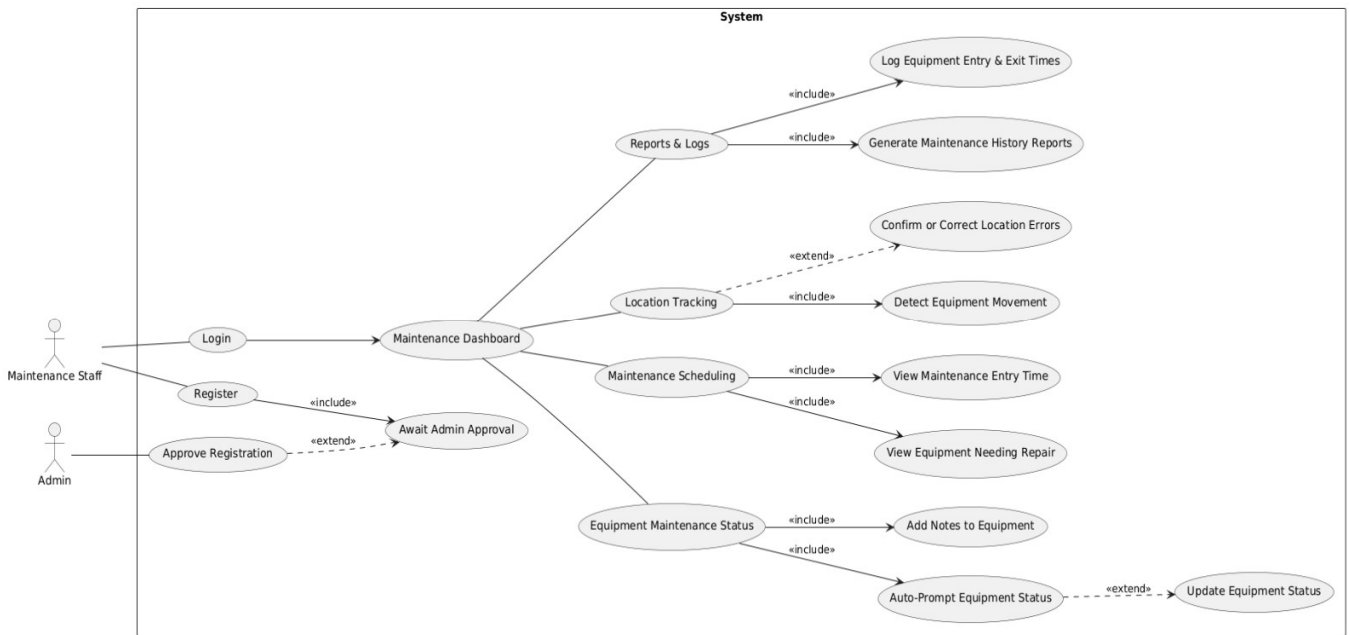


Figure 4.6.1.3: Maintenance Staff Use Case Diagram

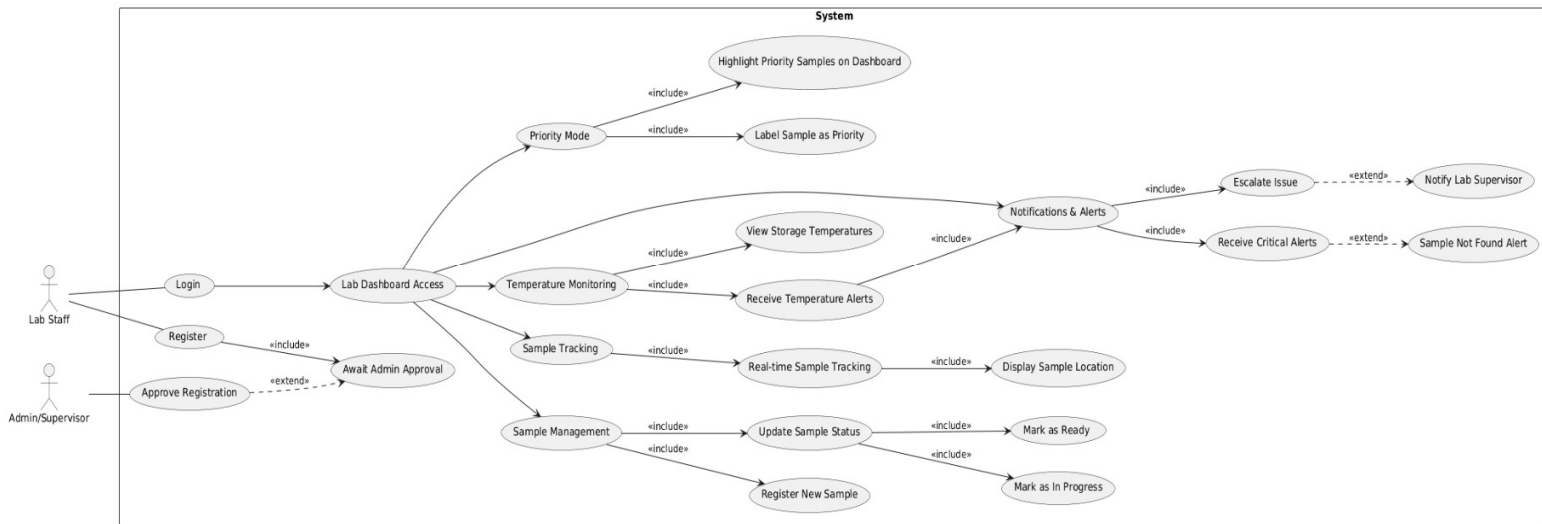


Figure 4.6.1.4: Laboratory Staff Use Case Diagram

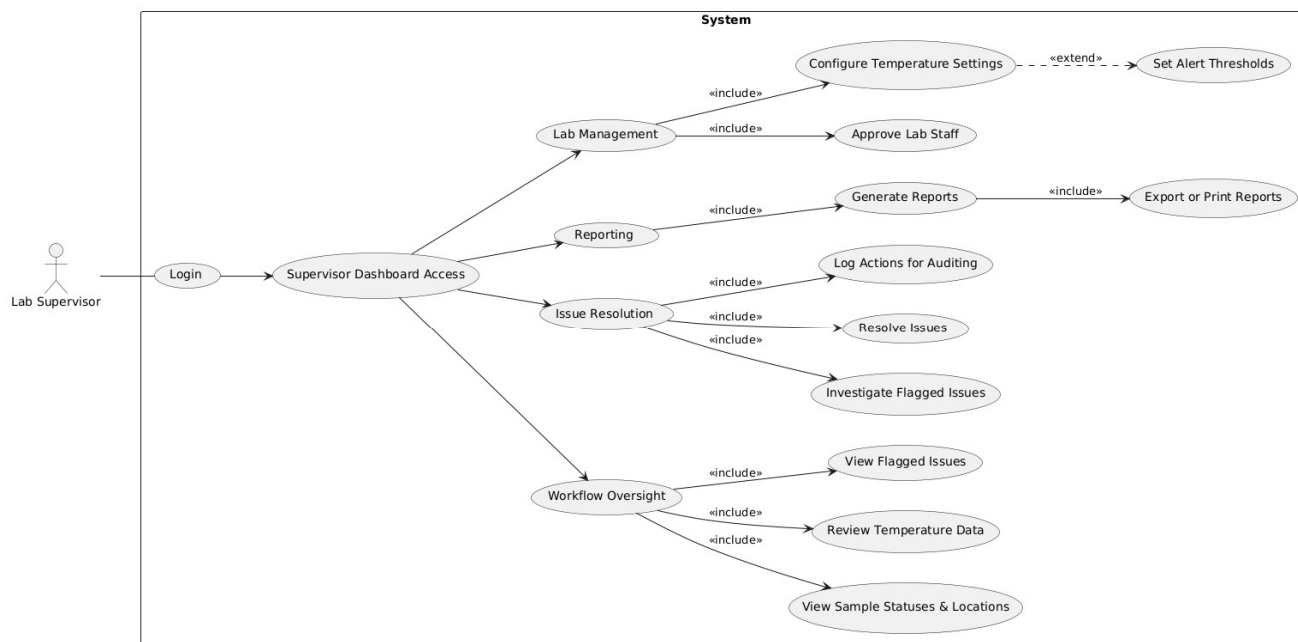


Figure 4.6.1.5: Laboratory Supervisor Use Case Diagram

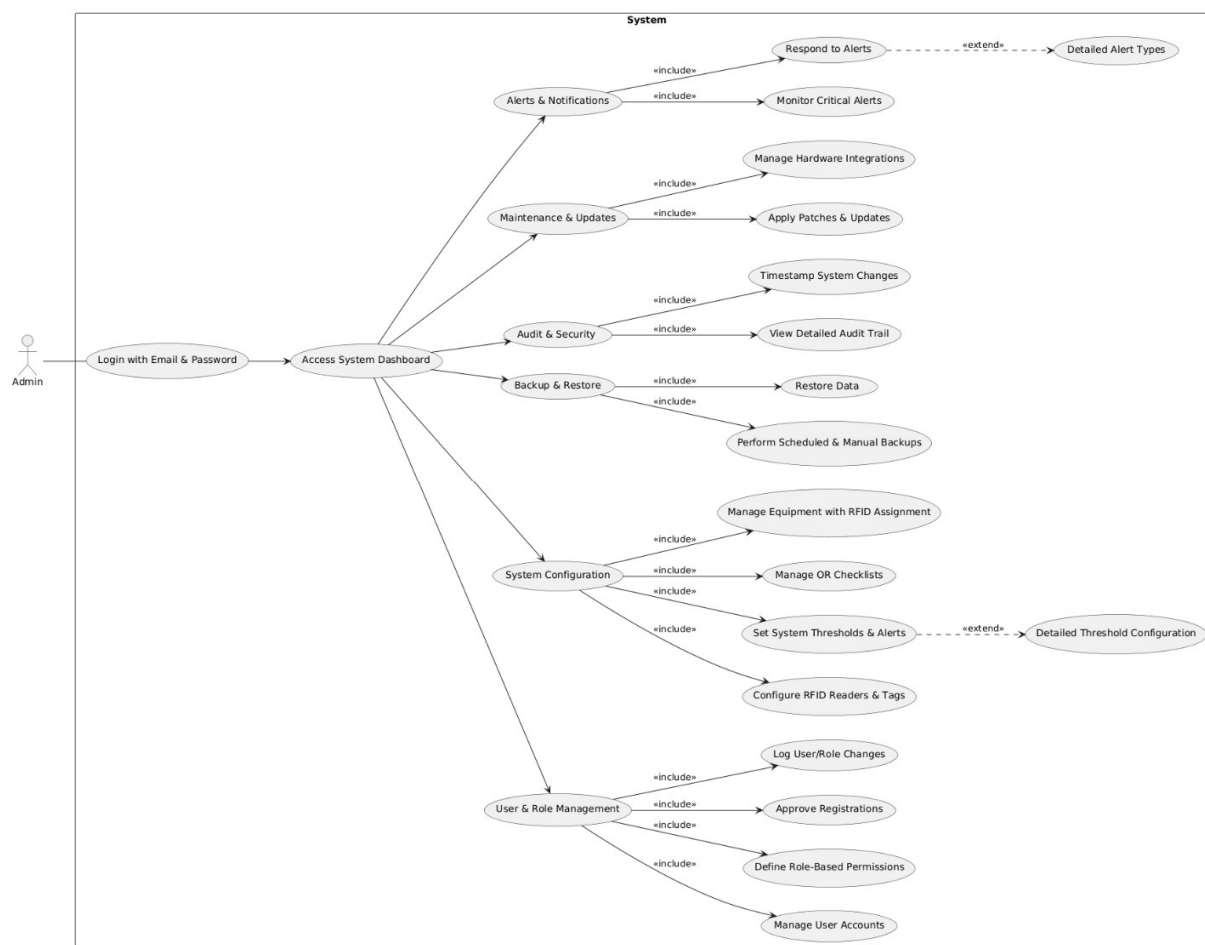


Figure 4.6.1.6: Admin Use Case Diagram

5. Chapter Four: System Design

5.1 Introduction

This chapter outlines the system design and the implementation specifics of the proposed Smart Health Care System. Using the user stories, scenarios, and requirements from the previous chapter, we create a clear technical plan. The aim is to show how each system component—Operating Room (OR), Sterilization, Maintenance, and Laboratory—works together to provide a cohesive, real-time workflow management platform driven by RFID technology.

5.2 Sequence Diagrams

(See the next page)



Figure 5.2.1 : OR staff Sequence Diagram

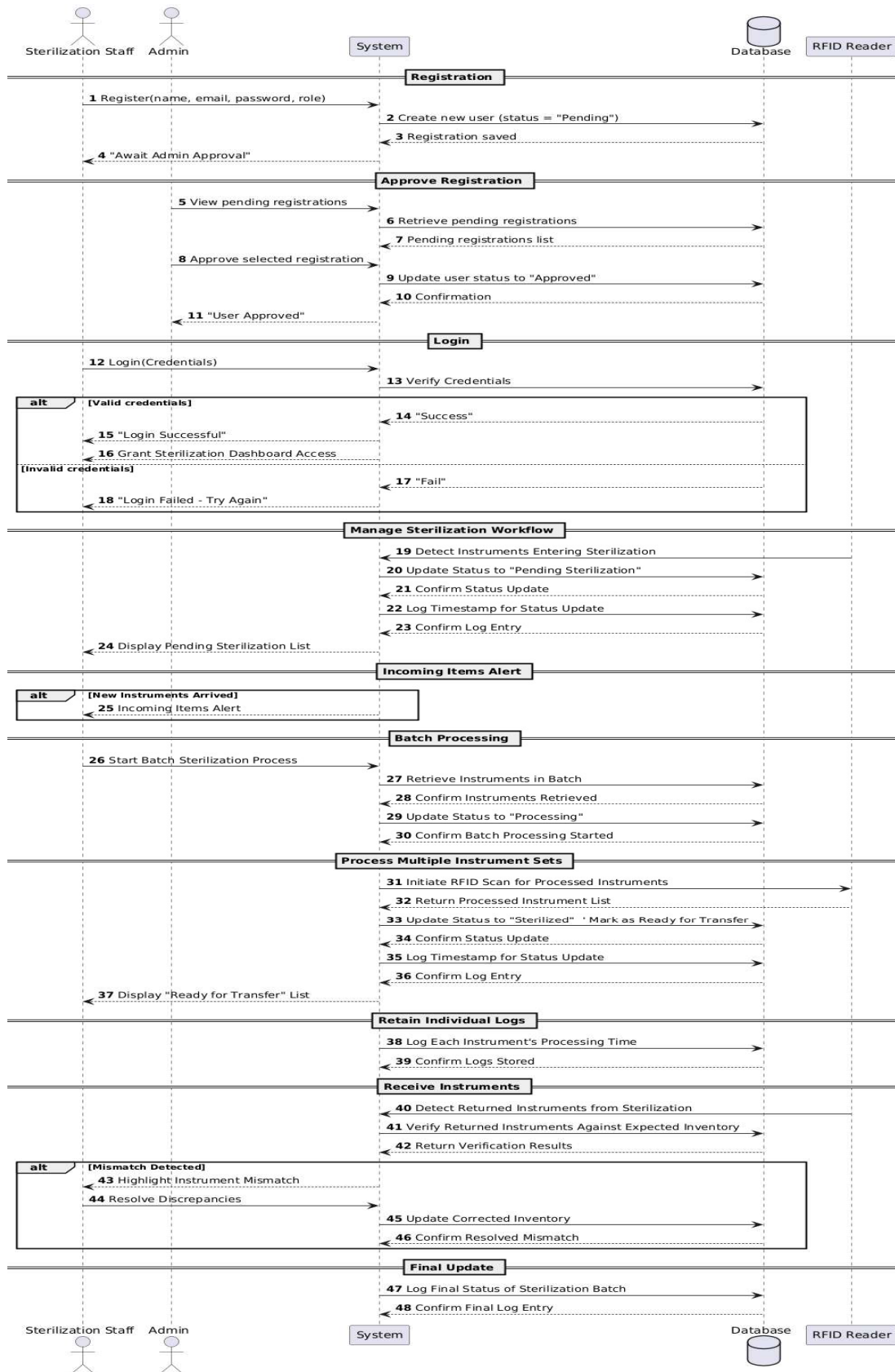


Figure 5.2.2 : Sterilization staff Sequence Diagram

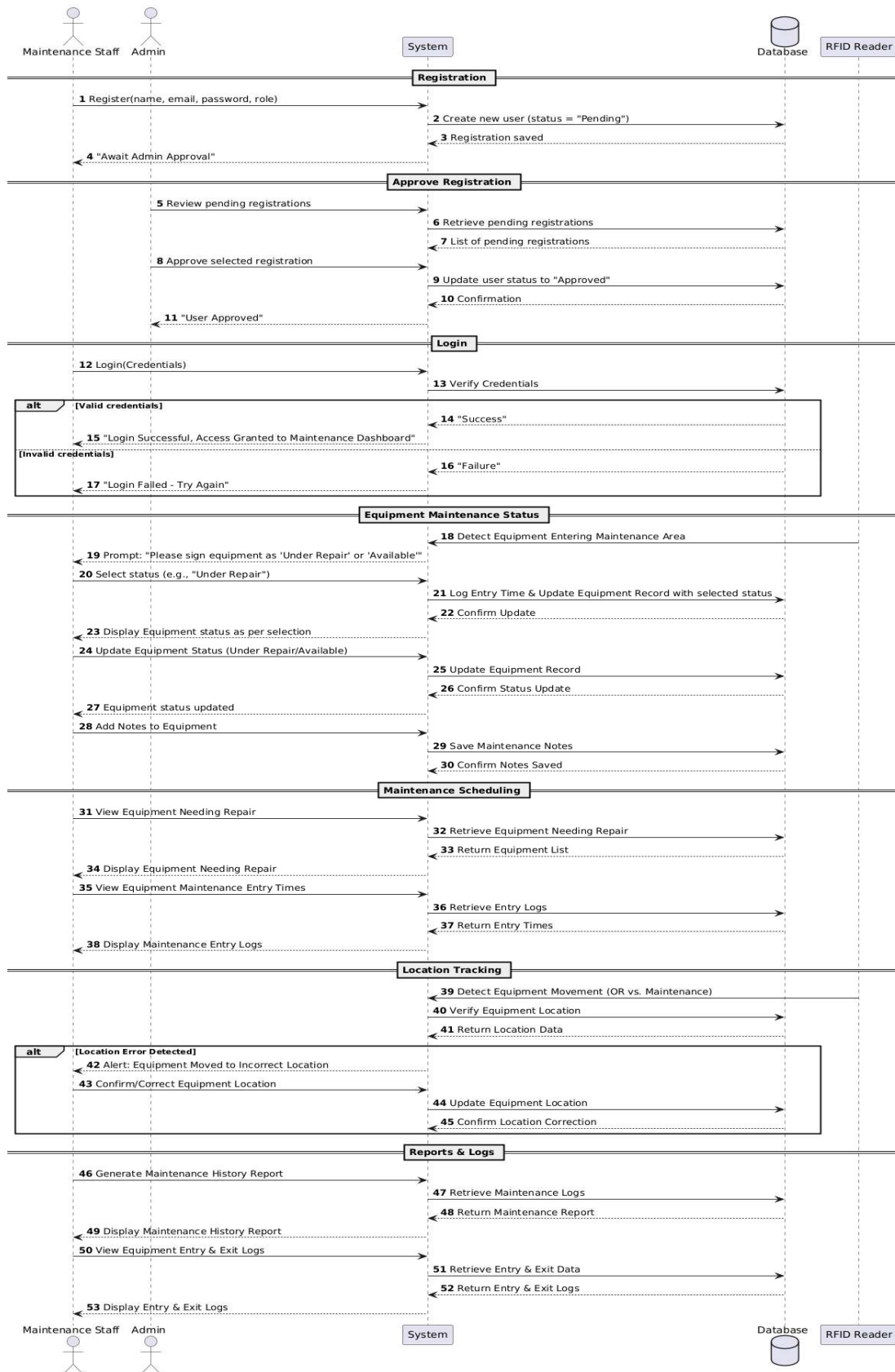


Figure 5.2.3 : Maintenance staff Sequence Diagram

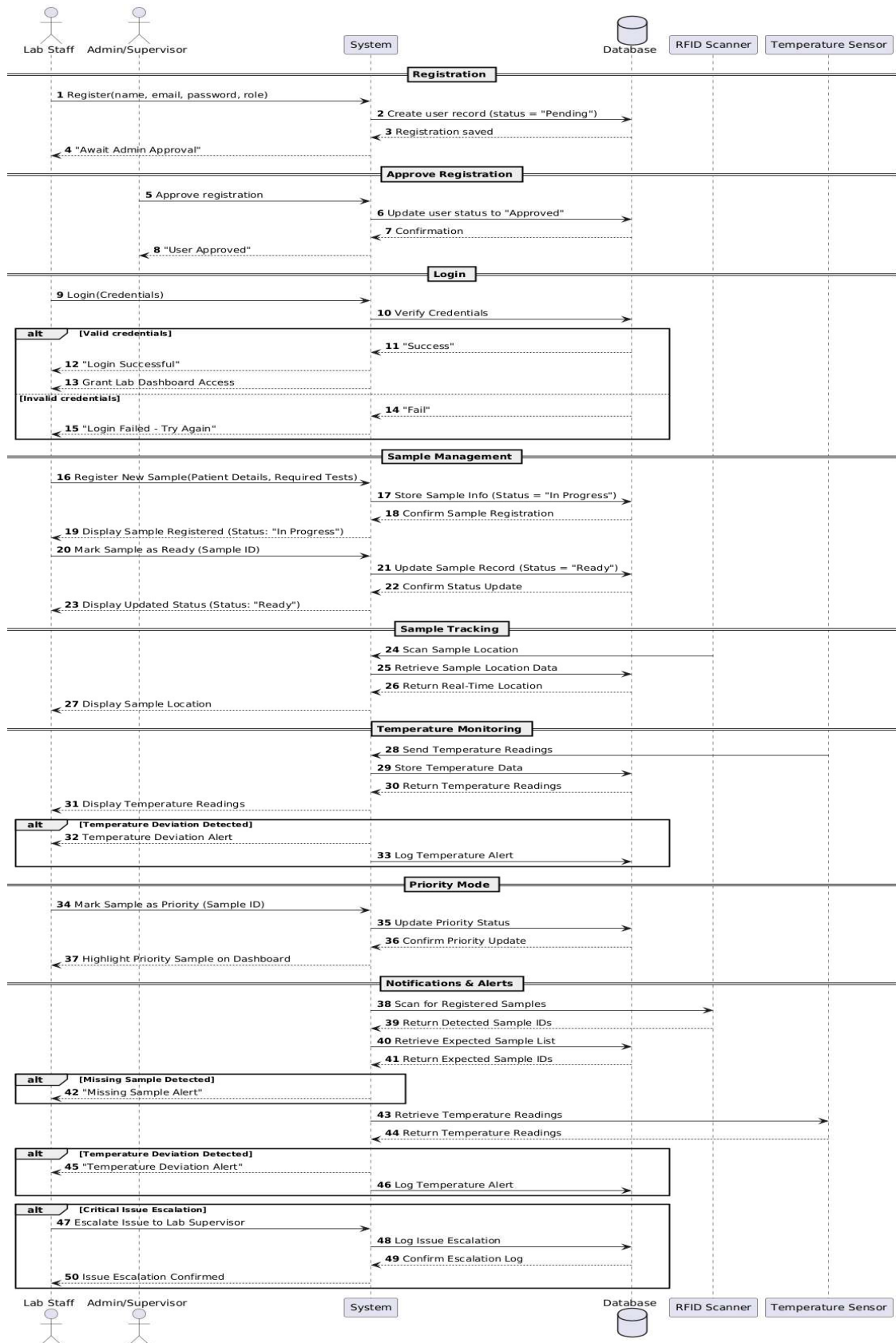


Figure 5.2.4 : Laboratory Staff Sequence Diagram

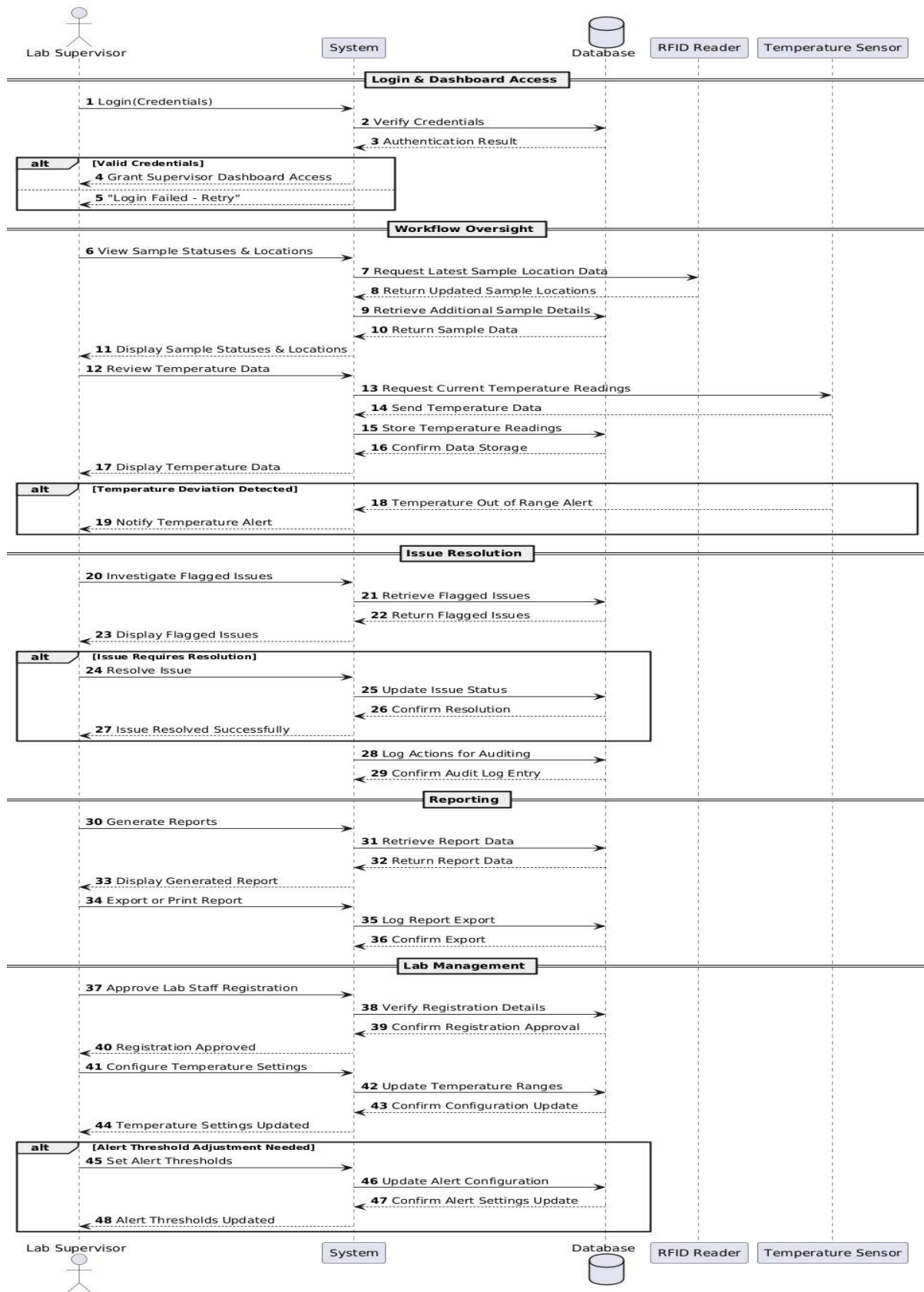


Figure 5.2.5 : Laboratory Supervisor Sequence Diagram

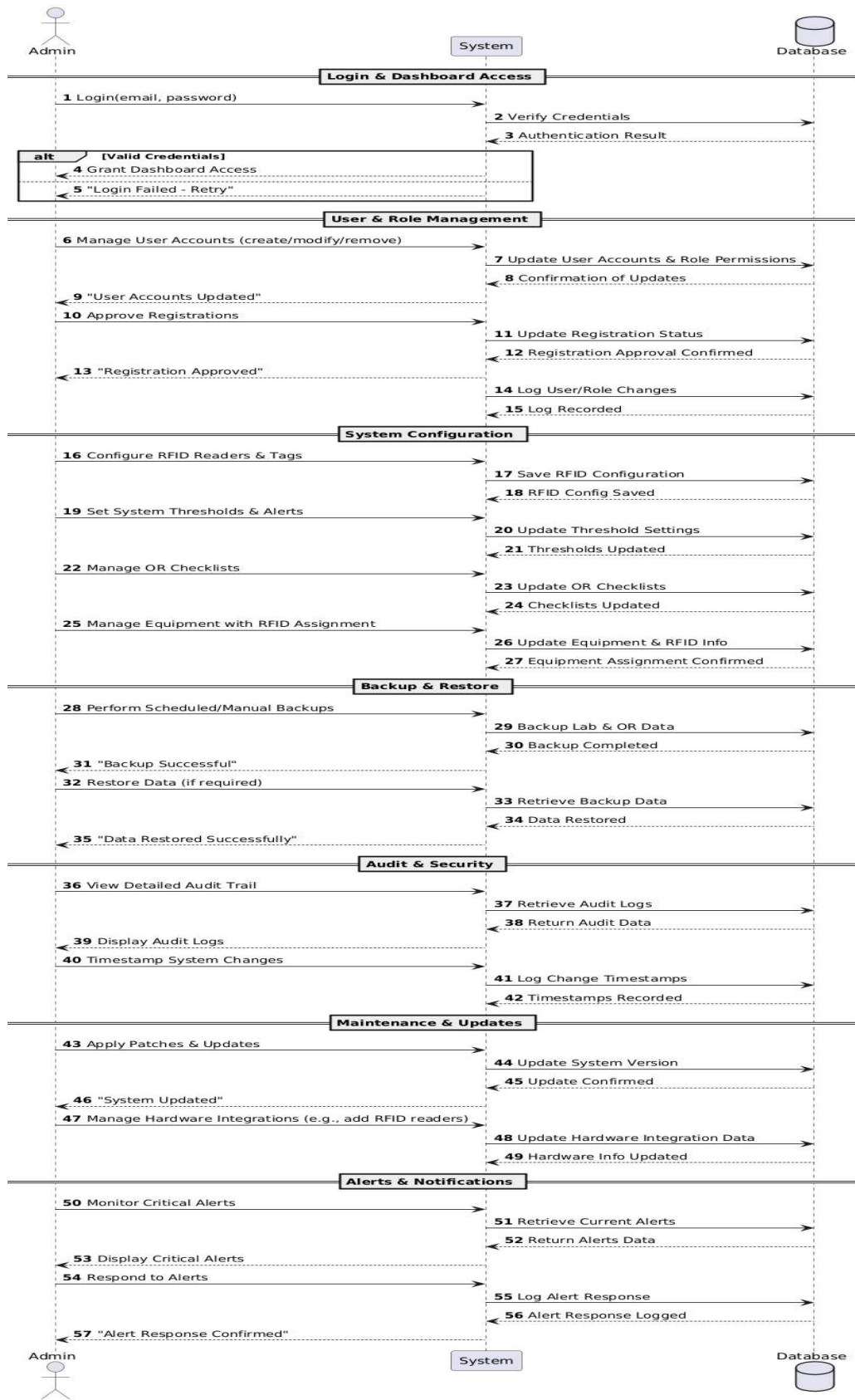


Figure 5.2.6 : Admin Sequence Diagram

5.3 Class Diagrams

(See the next page)

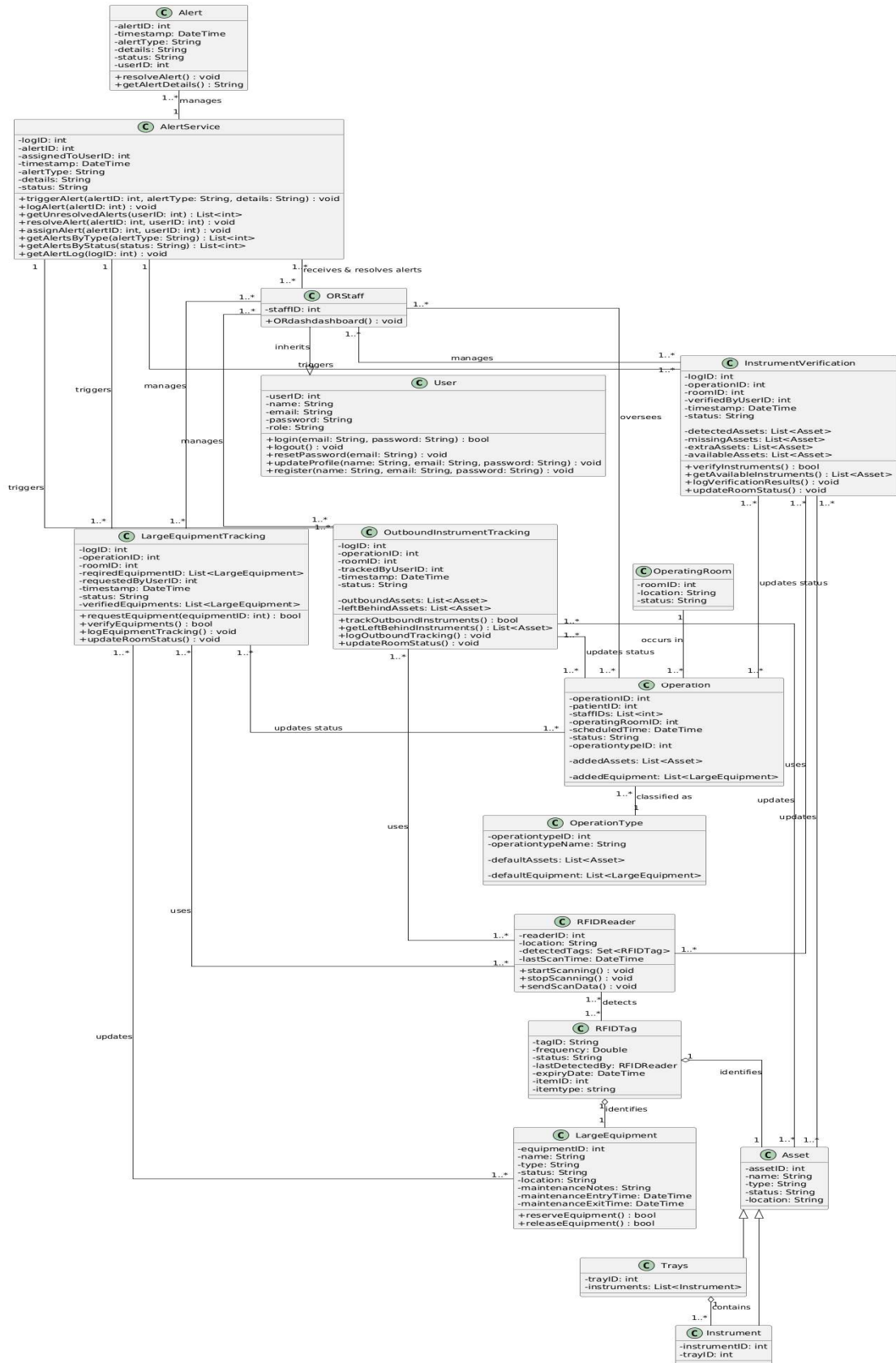


Figure 5.3.1: OR Staff Class Diagram

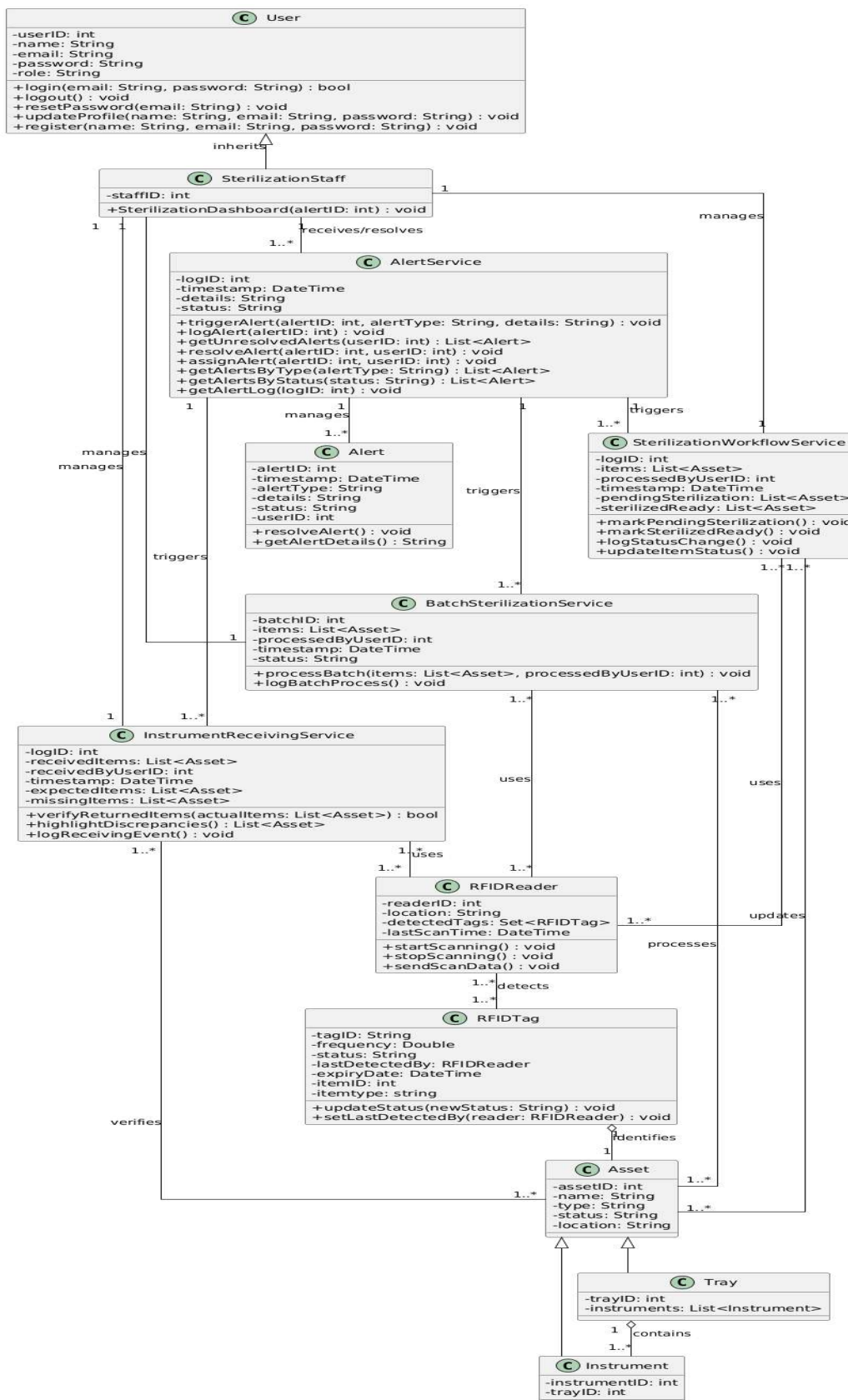


Figure 5.3.2: Sterilization Staff Class Diagram

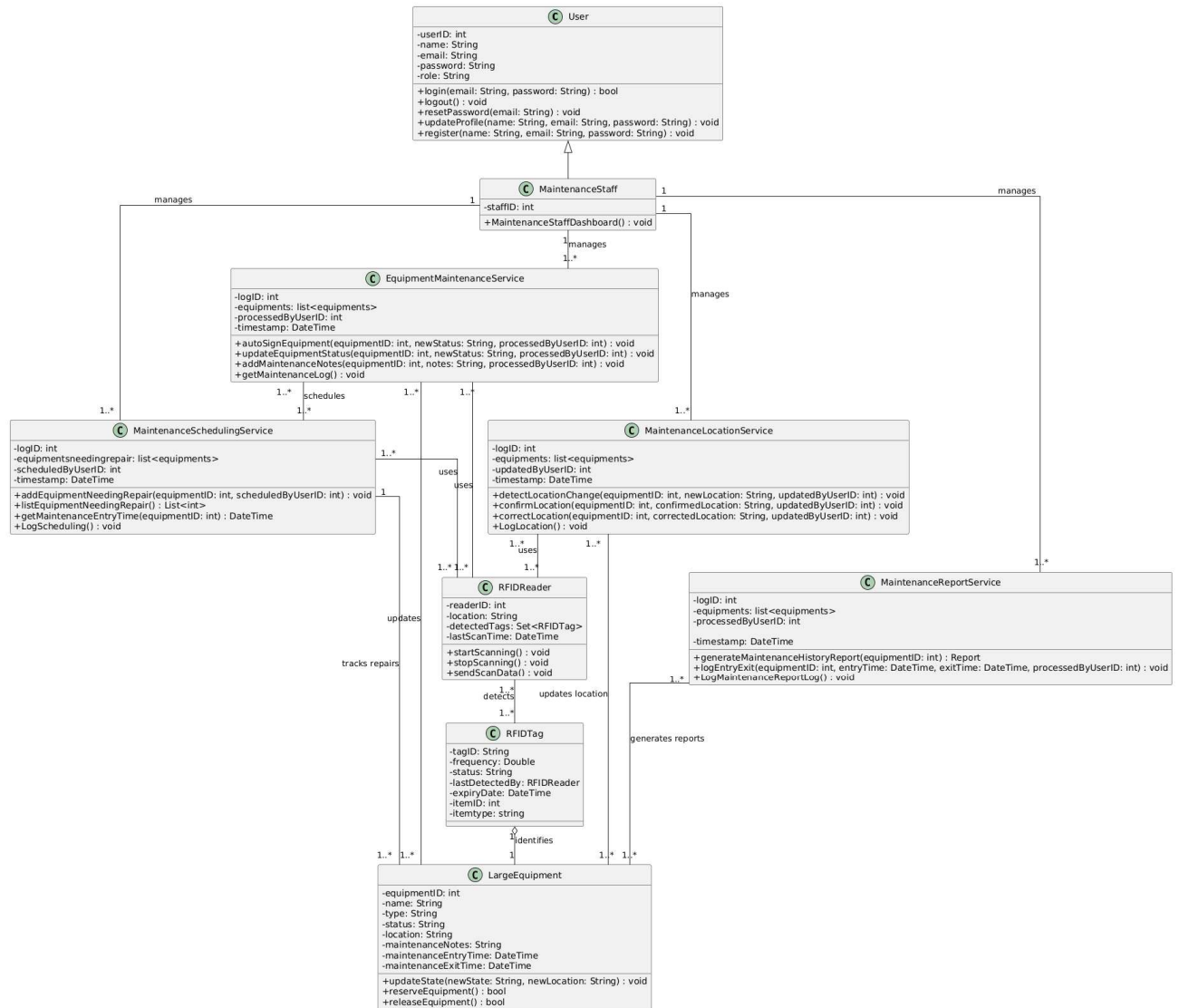


Figure 5.3.3: Maintenance Staff Class Diagram

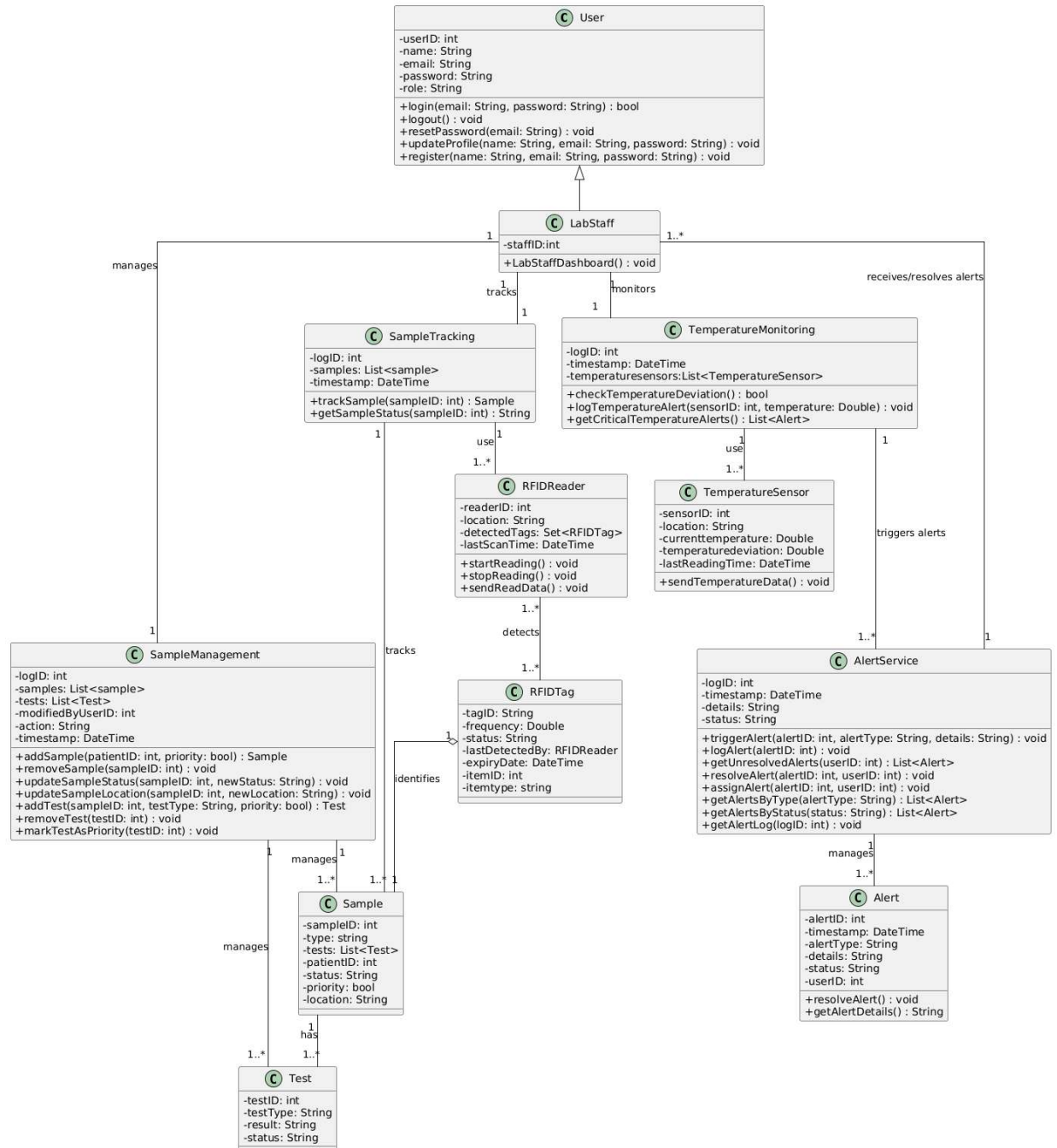


Figure 5.3.4: Laboratory Staff Class Diagram

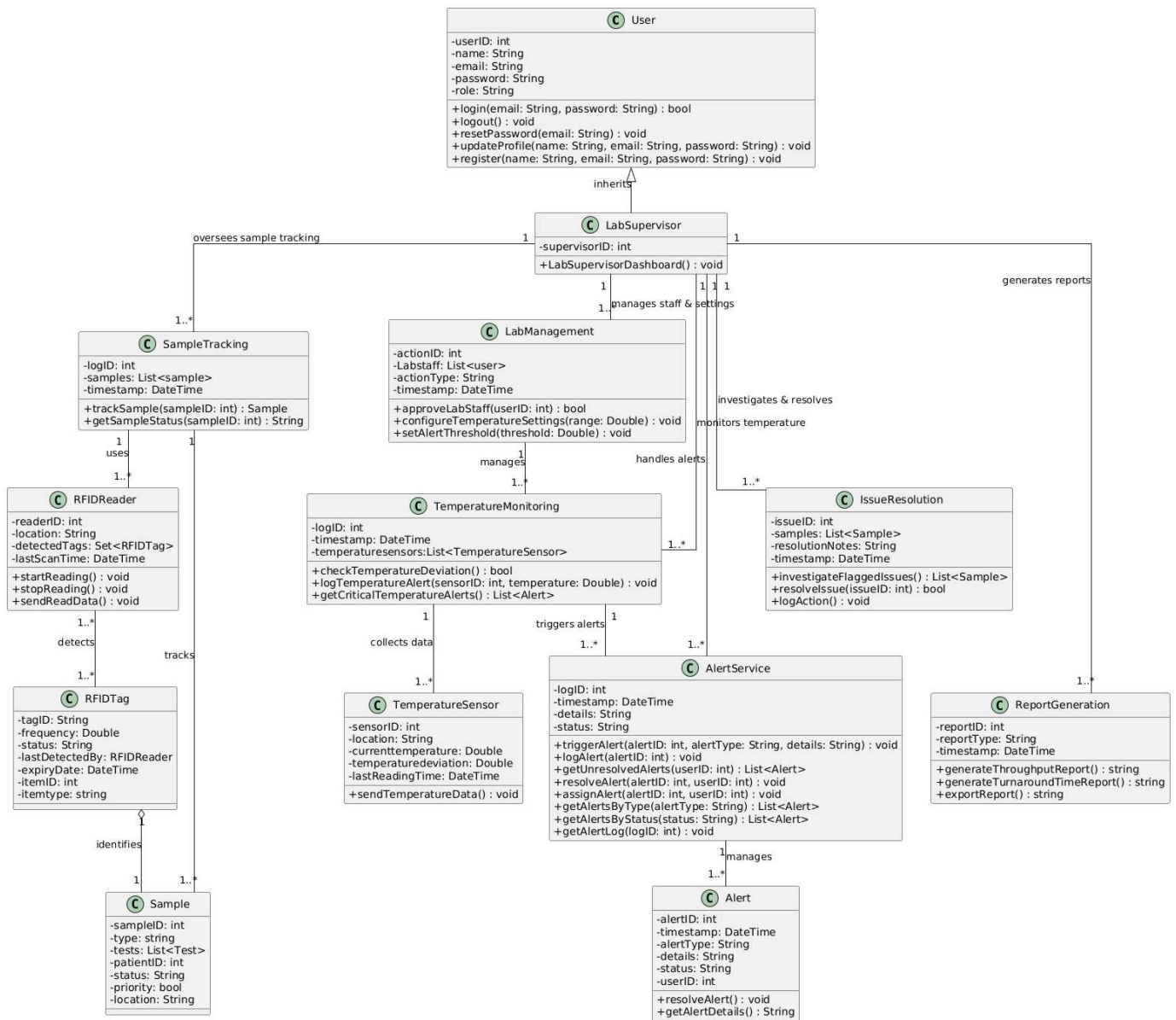


Figure 5.3.5: Laboratory Supervisor Class Diagram

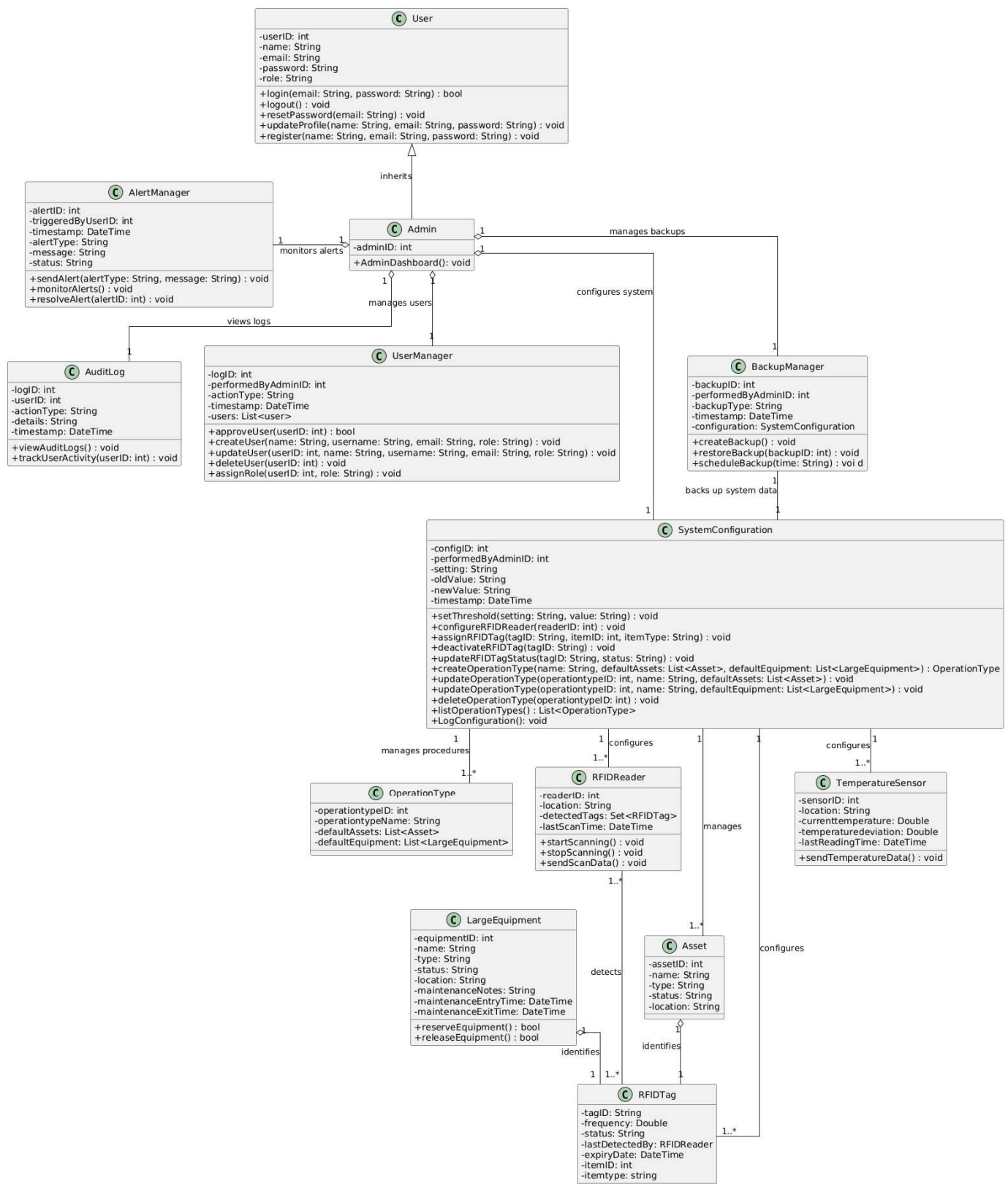


Figure 5.3.6: Admin Staff Class Diagram

5.4 ERD Diagrams

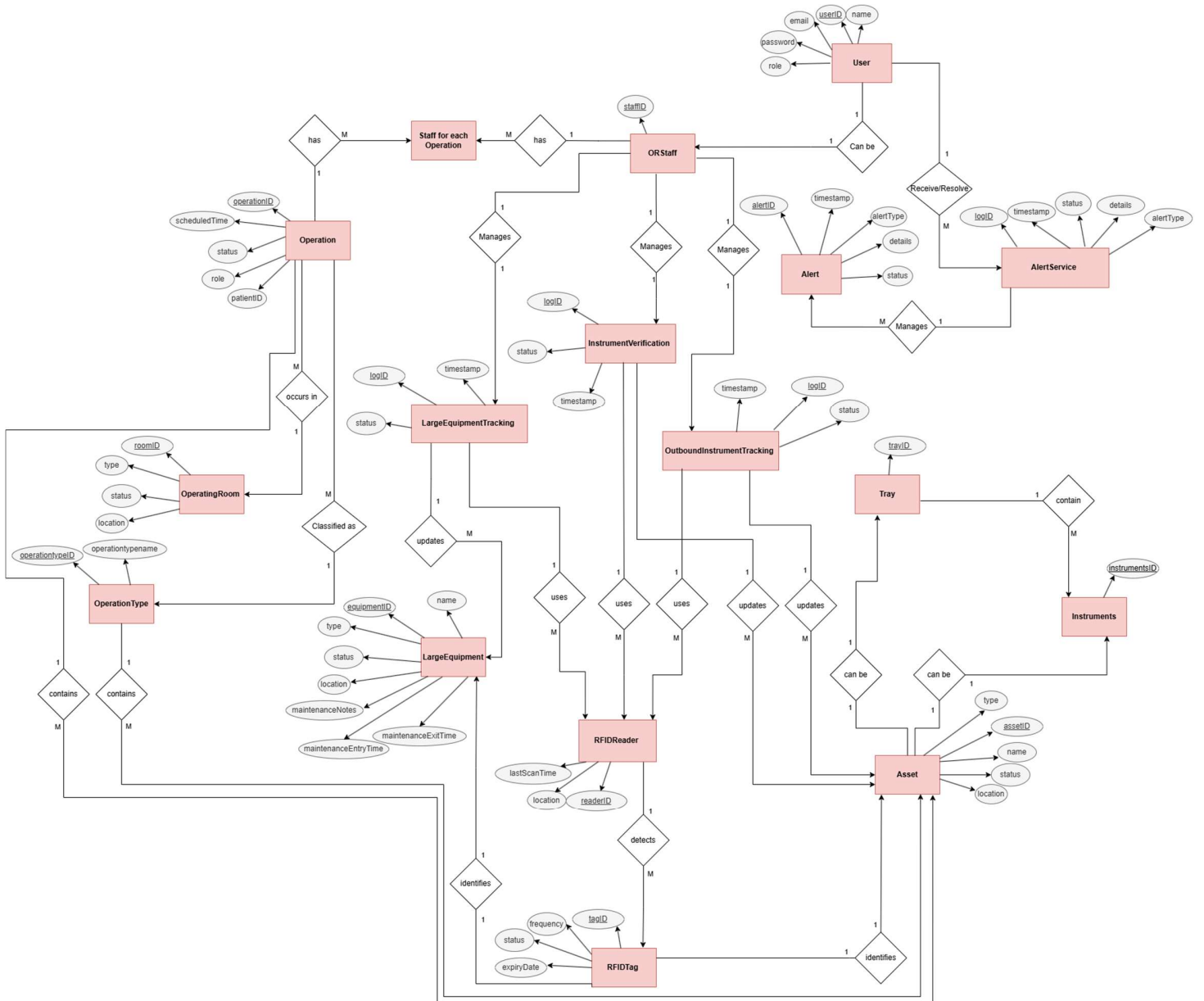


Figure 5.4.1: OR Staff ERD Diagram



Figure 5.4.2: Sterilization Staff ERD Diagram

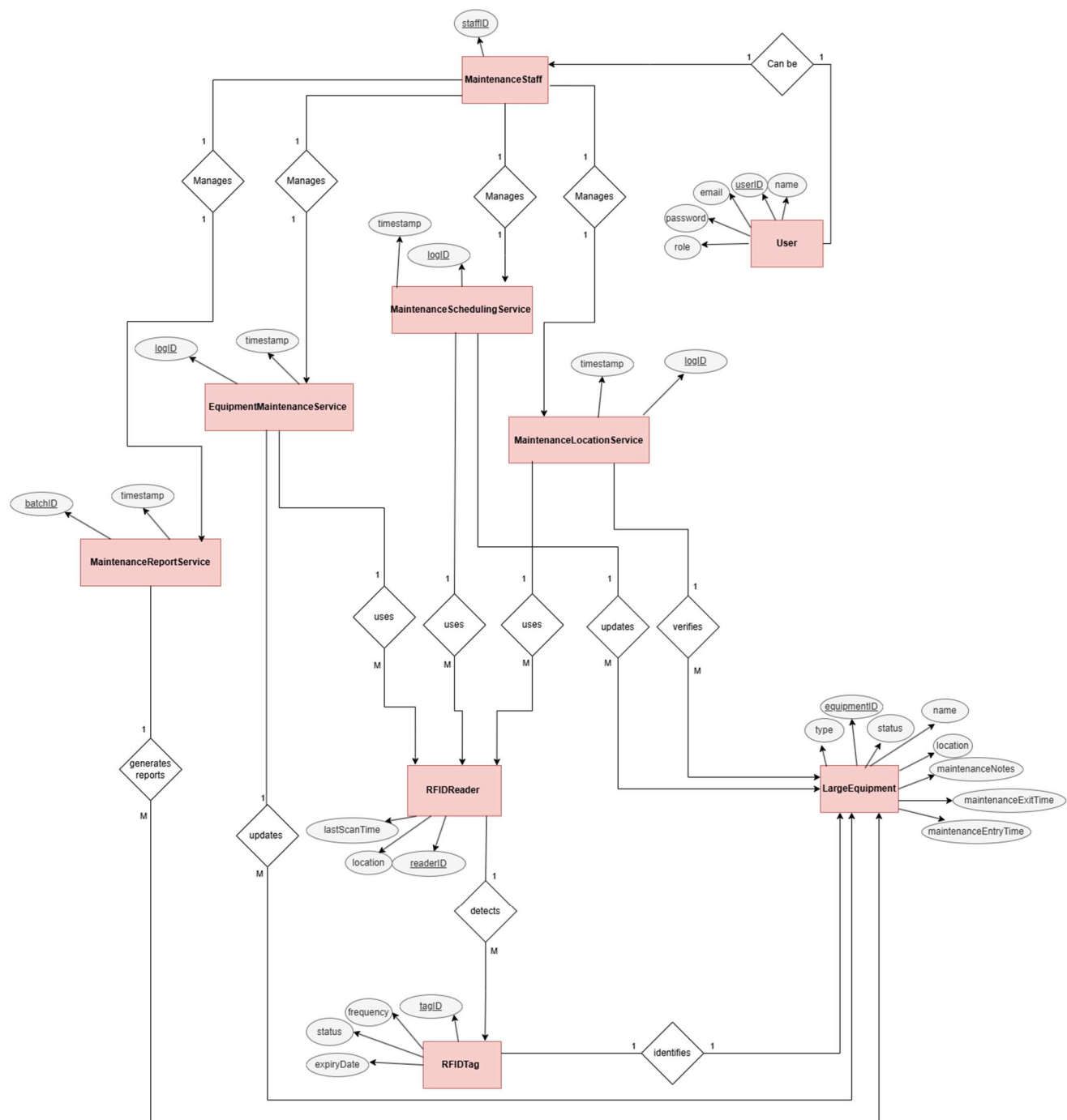


Figure 5.4.3: Maintenance Staff ERD Diagram

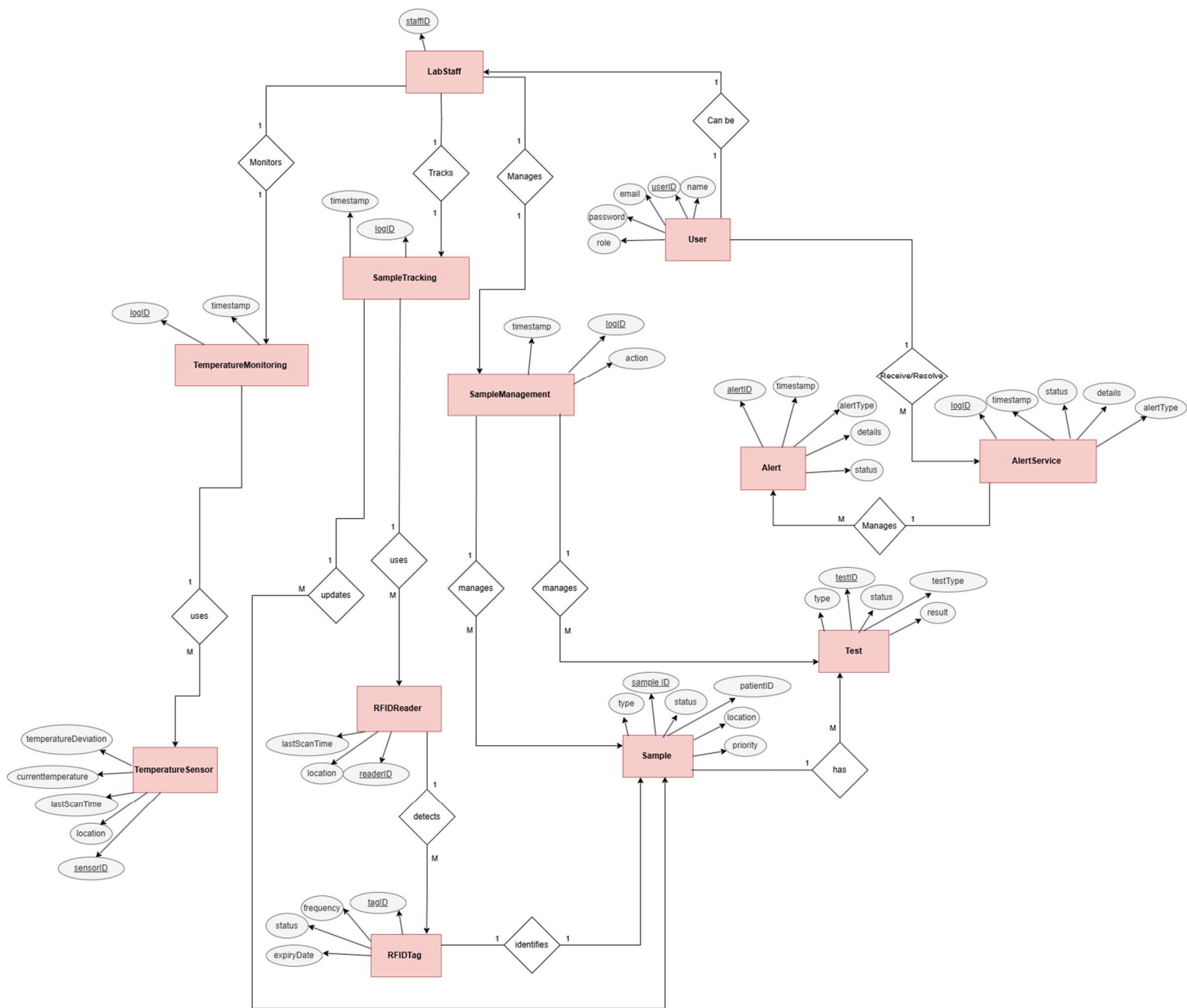


Figure 5.4.4: Laboratory Staff ERD Diagram

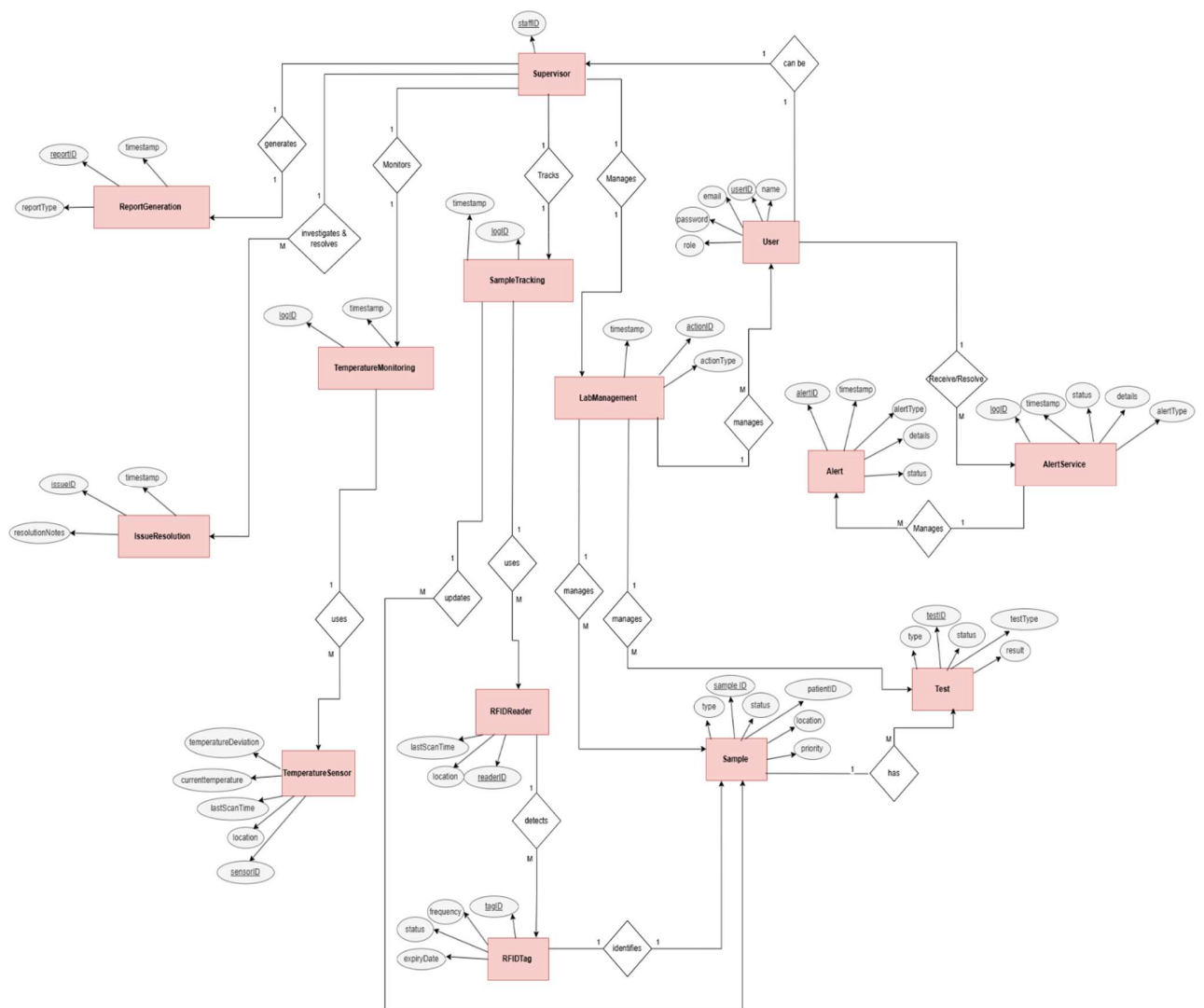


Figure 5.4.5: Laboratory Supervisor Staff ERD Diagram

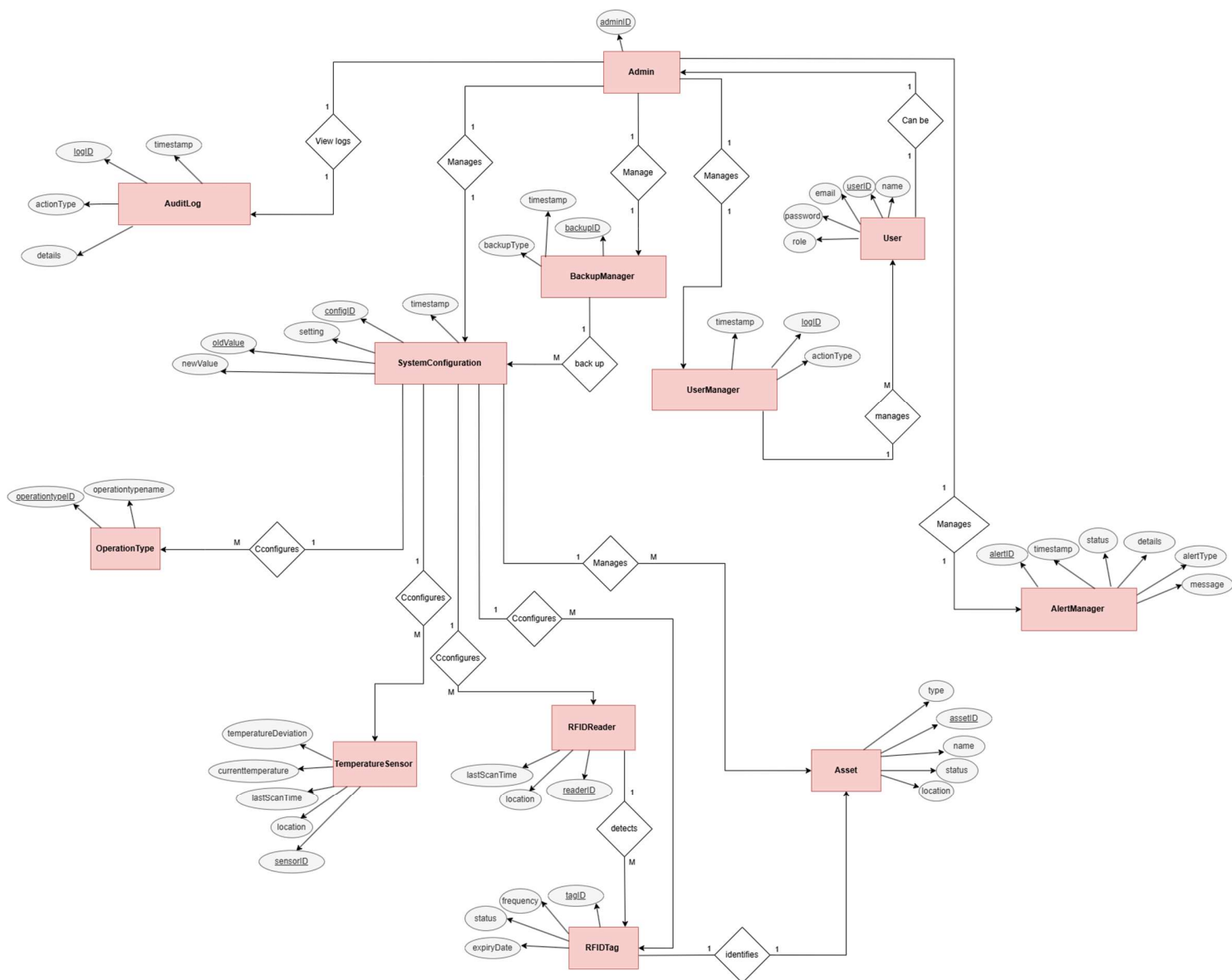


Figure 5.4.6: Admin Staff ERD Diagram

5.5 Context Design

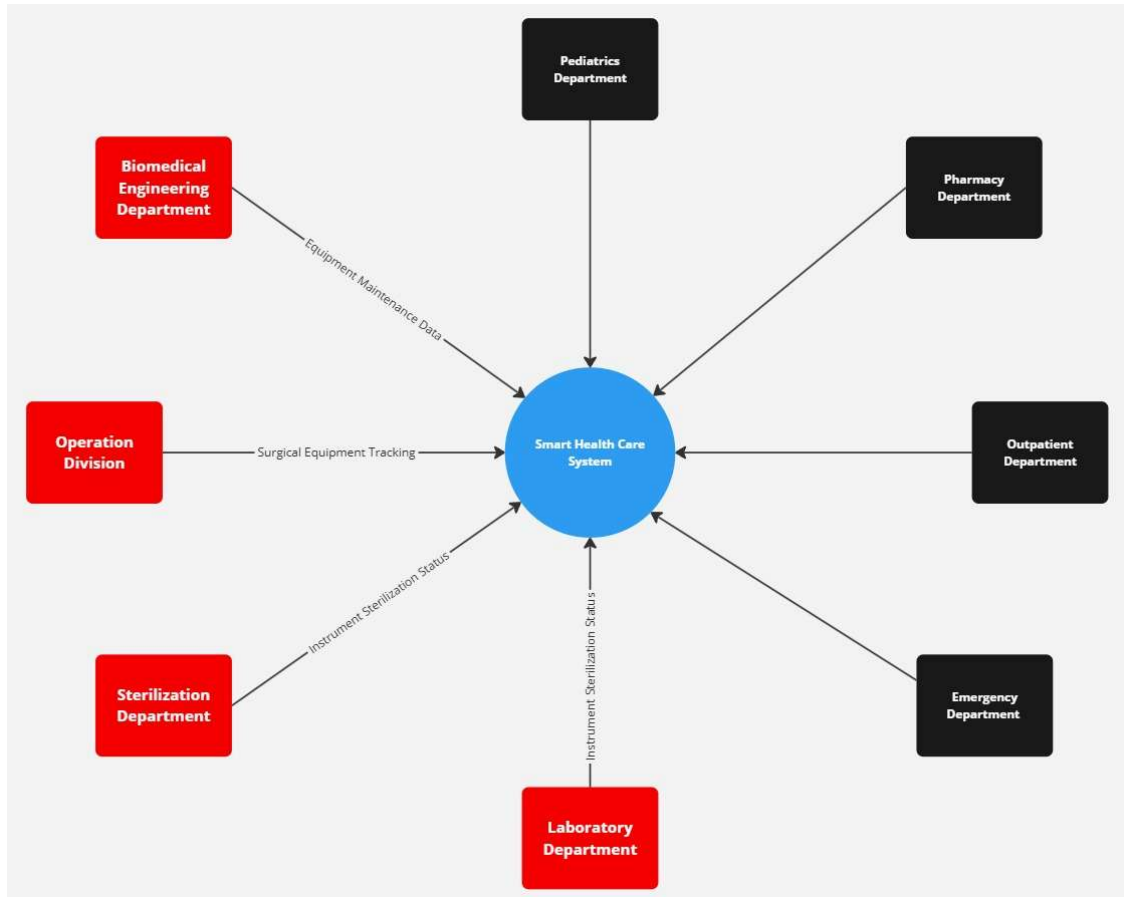


Figure 5.5.1: Context Diagram

5.6 Interface Design



Login

Username

Password

[Forgot Password ?](#)

[Login](#)

Don't have an account? [Sign Up](#)

Register

Full Name

Username

Email address

Password

Department

[Register](#)

OR

Have an account ? [Sign In](#)

Figure 1.1 : Log in and Register Interface

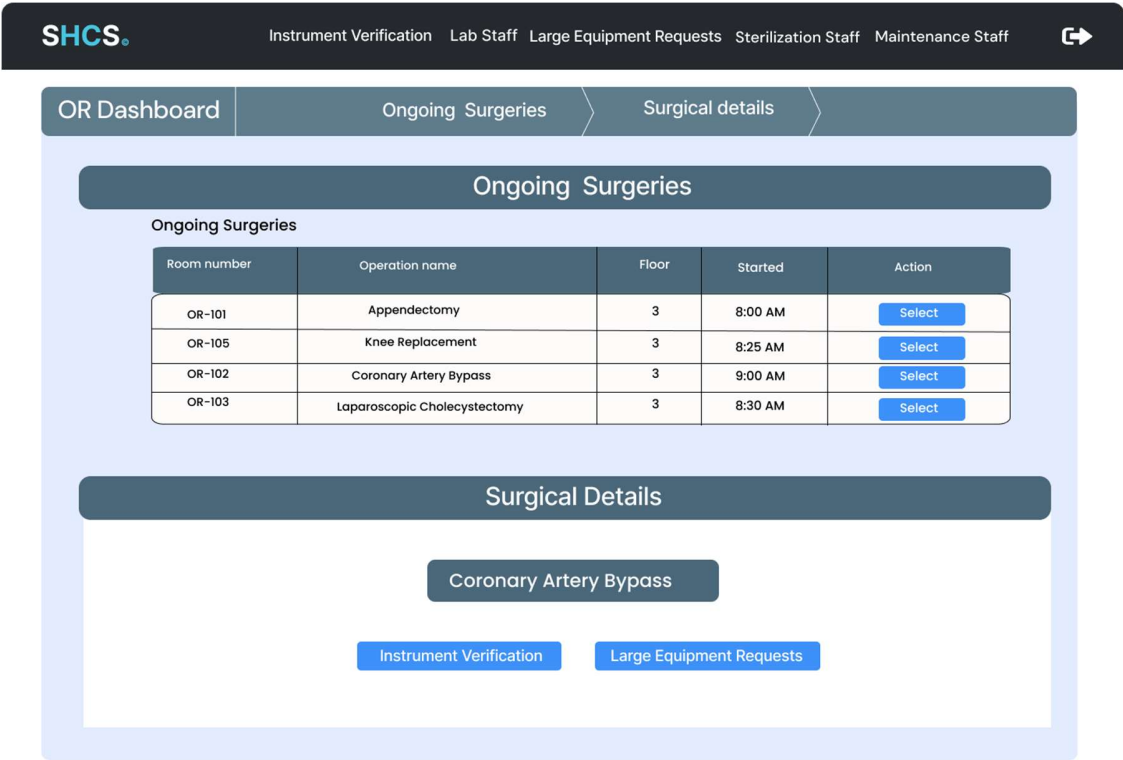


Figure 1.2 : OR Dashboard Interface

SHCS

Instrument Verification Lab Staff Large Equipment Requests Sterilization Staff Maintenance Staff

Instrument Verification

Required Instruments

missing

Available

Coronary Artery Bypass

All required instruments are ready and verified.

Mismatch detected! Please review the instruments.

Required Instruments :

NO.	Instrument ID	Item Name	quantity	Pre-use inside OR
1	001	Scalpel	1	☒
2	110	Sponge	2	☒
3	020	Retractor (Small)	2	☒

Add

Remove

missing :

NO.	Instrument ID	Item Name	quantity	State	location
1	011	Forceps (Tissue)	1	Missing	Storage Room A
2	021	Needle Driver	1	Damaged	Sterilization
3	031	Scissors	1	Missing	OR 3

Request a replacement tool

Available :

NO.	Instrument ID	Item Name	quantity	State	location
1	111	Hemostat	12	Sterile	OR Supply 1
2	211	Towel Clamp	6	Sterile	OR Supply 2
3	221		1	Sterile	OR Supply 2

← Back

Ready

Figure 1.3 : Instrument Verification Interface

SHCS.

Instrument Verification Lab Staff Large Equipment Requests Sterilization Staff Maintenance Staff

Outbound Instrument Tracking

Required Instrument

Instruments to Leave

Departure Log

Coronary Artery Bypass

No instruments left behind.

Required instrument :

NO.	Instrument ID	Item Name	Quantity	Pre-use Inside OR
1	042	Scalpel	1	✓
2	333	Hemostat	4	✓
3	223	Needle Driver	1	✓

Request Devices

Instruments to Leave :

NO.	Instrument ID	Item Name	Departure Time	State	Location
1	021	Sponge	2025-02-08 10:15 AM	Departed	Central Sterile
2	026	Retractor (Medium)	2025-02-08 10:22 AM	Left Behind	Sterilization
3	056	Needle Driver		Departed	Central Sterile

Departure Log :

NO.	Instrument ID	Item Name	Departure Time	State	Location
1	003	Sponge	10:30 AM	Departed	Sterilization
2	103	Scalpel	10:32 AM	Departed	Sterilization
3	013	Sponge	10:35 AM	Departed	Sterilization

← Back

Ready

Figure 1.4 : Outbound Instrument Tracking Interface

SHCS.

Instrument VerificationLab StaffLarge Equipment RequestsSterilization StaffMaintenance Staff

Large Equipment Requests

Availability

Required Equipment

Coronary Artery Bypass

The request was successful

Availability :

NO.	Code	Item Name	Departure Time	State	Location
1	465	X-Ray Machine	10:35 AM	Available	Radiology Dept.
2	456	Fluoroscopy Machine (C-Arm)	11:00 AM	Available	Imaging Suite 2

View Equipment Location

Required Equipment :

NO.	Code	Item Name	Departure Time	State	Location
1	867	X-Ray Machine	11:30 AM	Available	Recovery Room 1
2	234	Patient Monitor	11:10 AM	Unavailable	Pre-Op Holding
3					

Request Equipment

Back

Process Batch

Figure 1.5 : Large Equipment Requests Interface

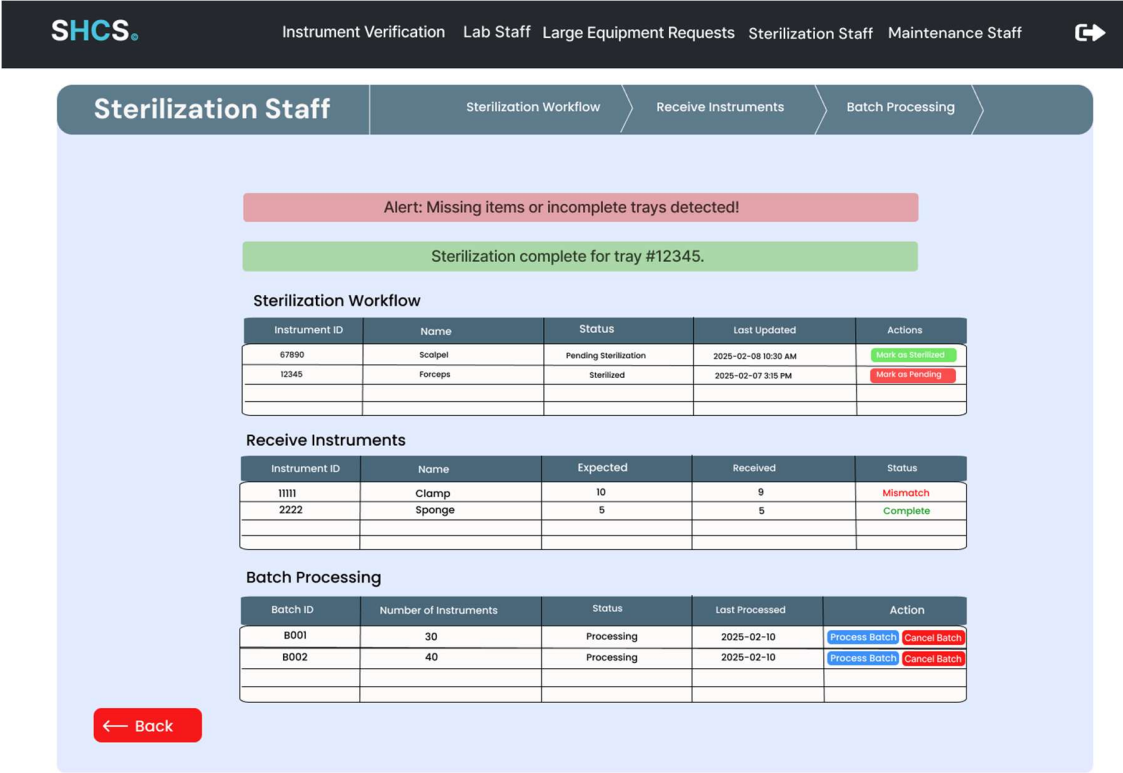


Figure 1.6 : Sterilization Staff Dashboard Interface

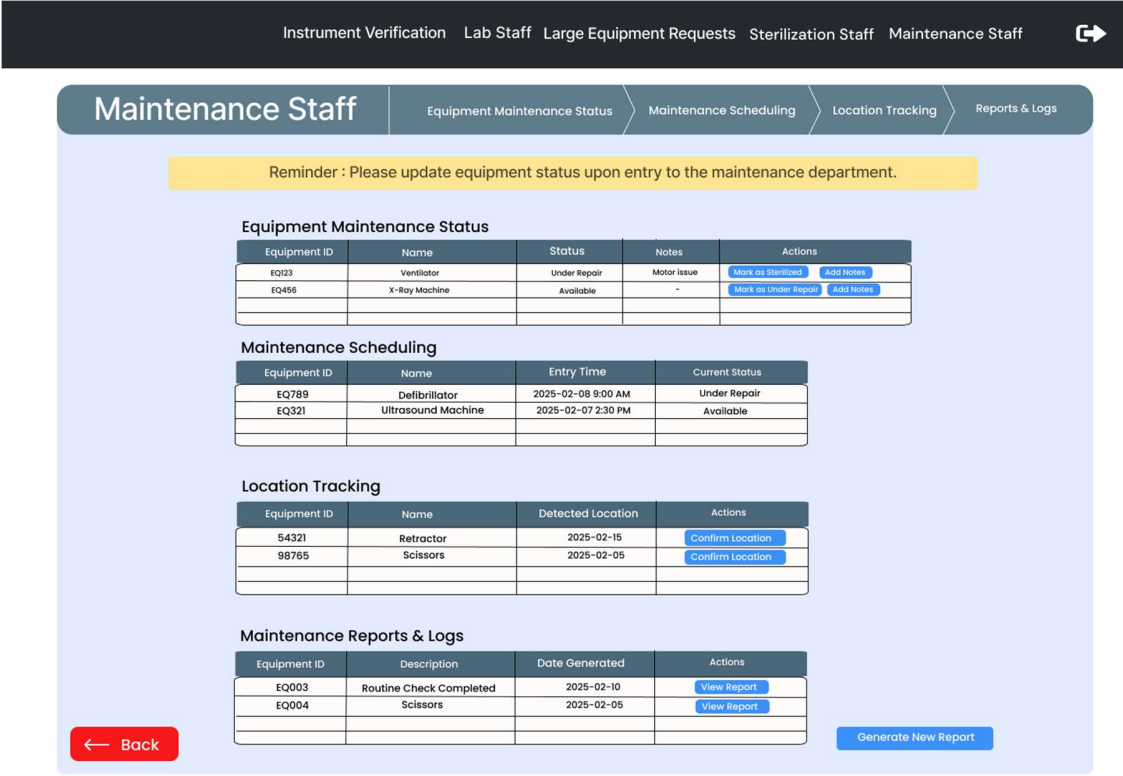


Figure 1.7 : Maintenance Staff Dashboard Interface



Figure 2.1 : Lab Staff Dashboard Interface

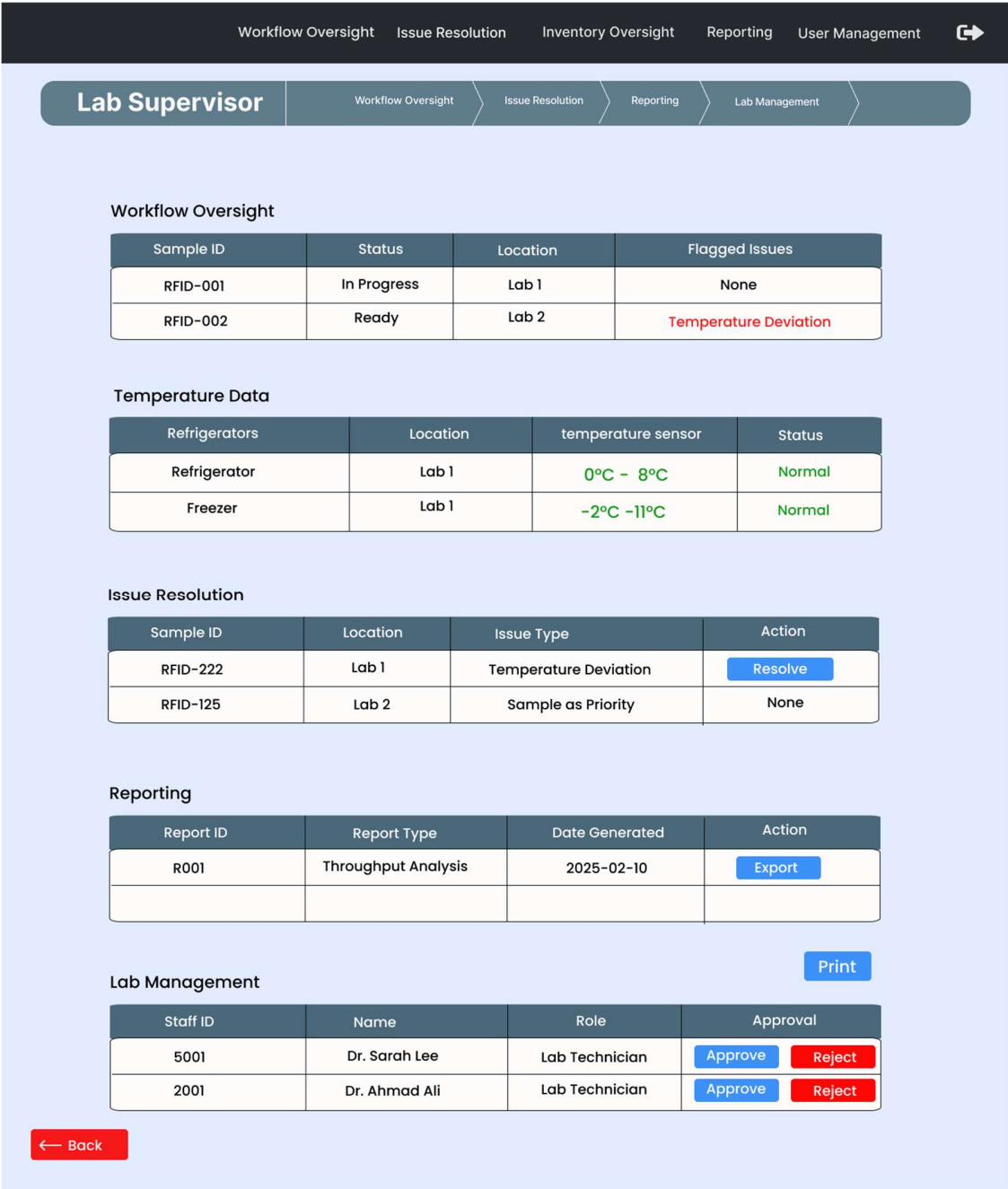


Figure 2.2 : Supervisor Dashboard Interface

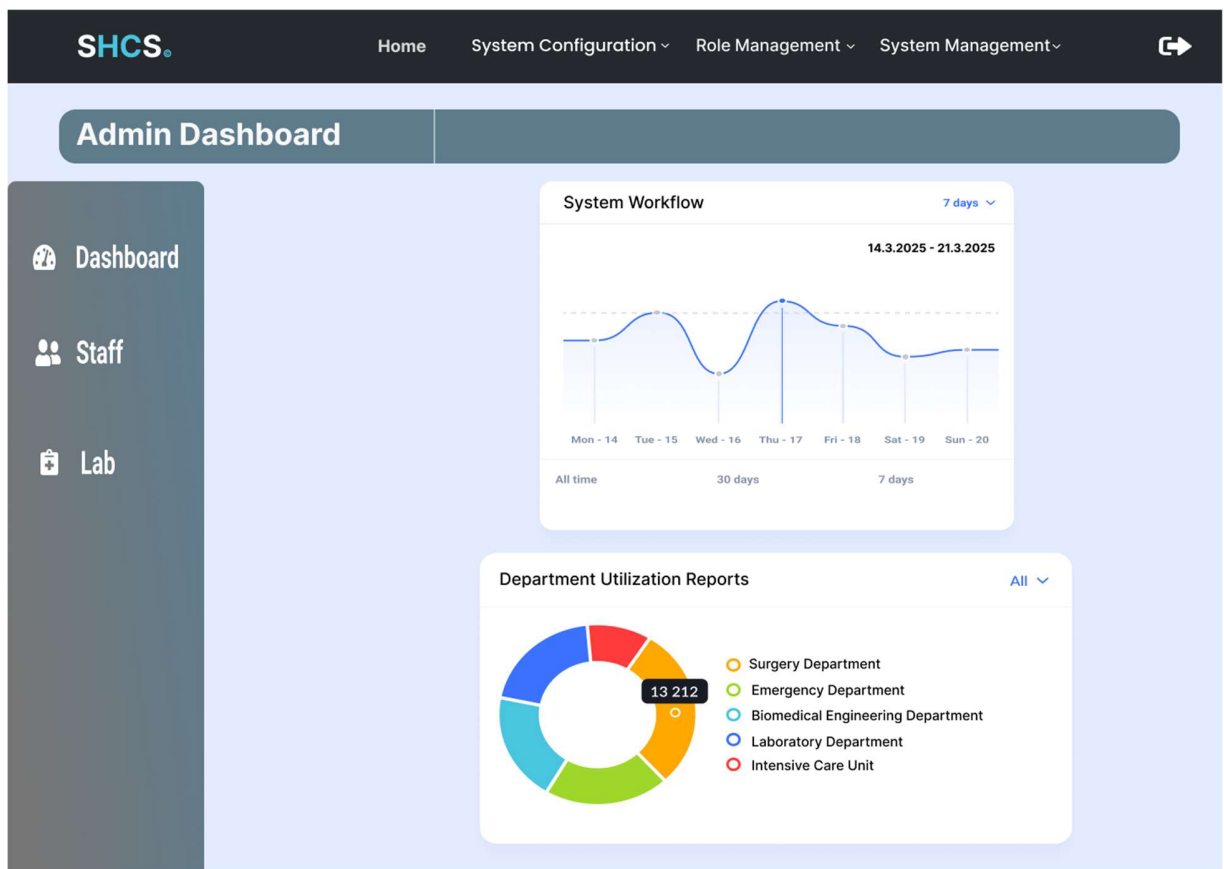


Figure 3.1 : Admin Dashboard Interface

SHCS

[Home](#) [System Configuration](#) [Role Management](#) [System Management](#)

User & Role Management

User & Role Management

Add/Edit User

User & Role Management

User ID	Name	Email	Role	Actions
001	Ali Hammade	john.doe@example.com	OR Staff	Edit Delete
002	joan Thaer	joan Thaer@example.com	OR Staff	Edit Delete

Add/Edit User

Full Name :

Email Address :

Password :

Role :

Save

[← Back](#)

Figure 3.2 : Admin User & Role Management Interface

SHCS

Home

System Configuration ▾

Role Management ▾

System Management ▾

↔

System Configuration

RFID Configuration

Alert Thresholds

Procedure Checklist

RFID Configuration

RFID Tag ID :

Assign to Instrument :

Assign RFID

Alert Thresholds

Set Tray Sterilization Alert Threshold :

Save Threshold

Procedure Checklist

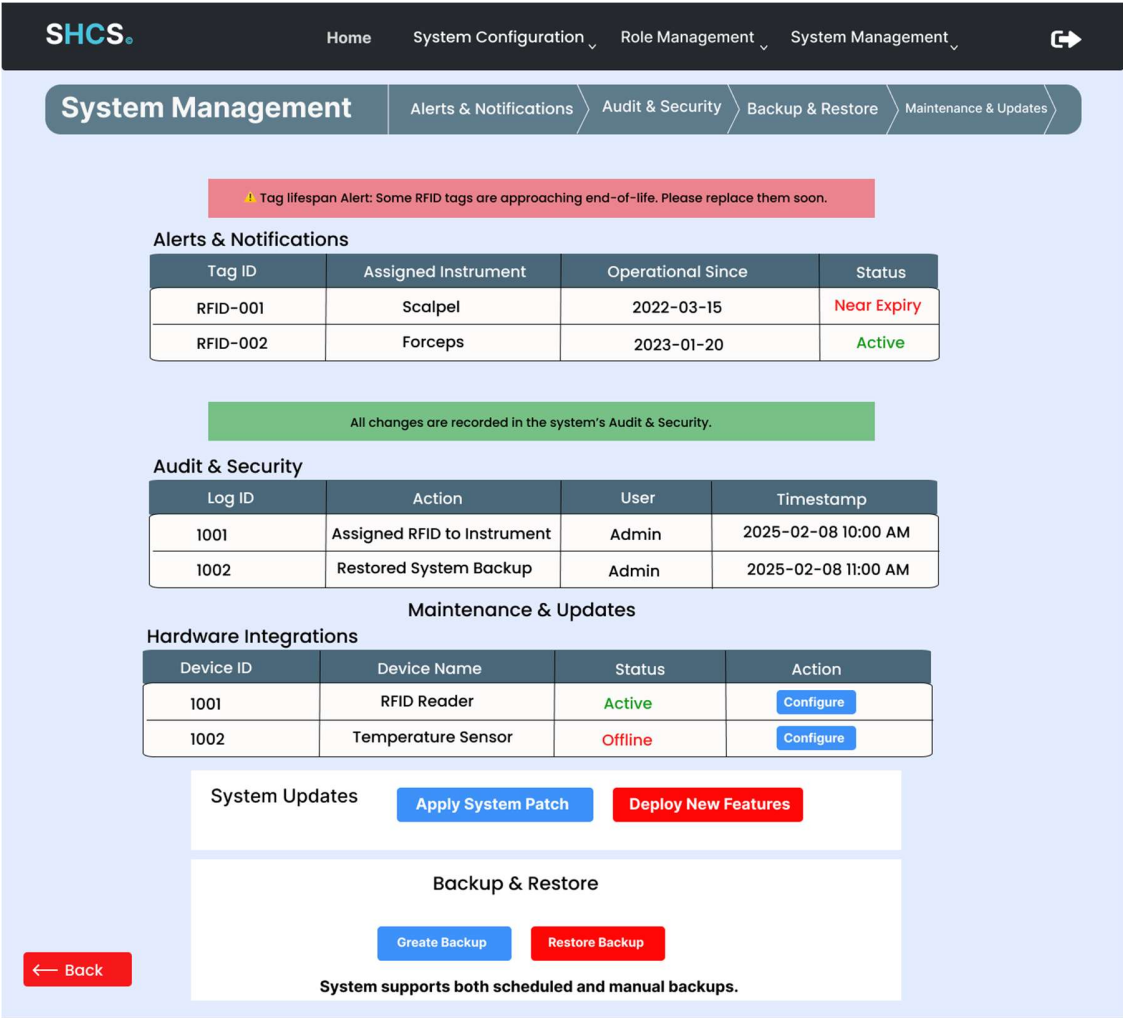
Operation Type:

Required Tools (Comma Separated) :

Save Checklist

← Back

Figure 3.3 : Admin System Configuration Interface



SHCS.

Home

System Configuration

User & Role Management

Security & Audit

Backup & Restore

Maintenance & Updates

System Configuration

Test Catalog Management

Device Integrations

Priority Rules

Test Catalog Management

Test ID	Test Name	Category	Action
T001	Complete Blood Count (CBC)	Hematology	Edit
T002	Glucose Test	Biochemistry	Edit

Add New Test

Device Integrations

Device ID	Device Name	Type	Status	Action
1001	RFID Scanner	RFID	Active	Configure
1002	Temperature Sensor	Environmental	Offline	Configure

Add New Device

Priority Rules

Rule ID	Description	Threshold	Action
P001	Mark samples as high priority if test is CBC	Auto-assign	Edit
P002	Send alert if temperature exceeds 8°C	8°C	Edit

Add New Rule

← Back

Figure 3.5 : Admin System Configuration Lab Interface

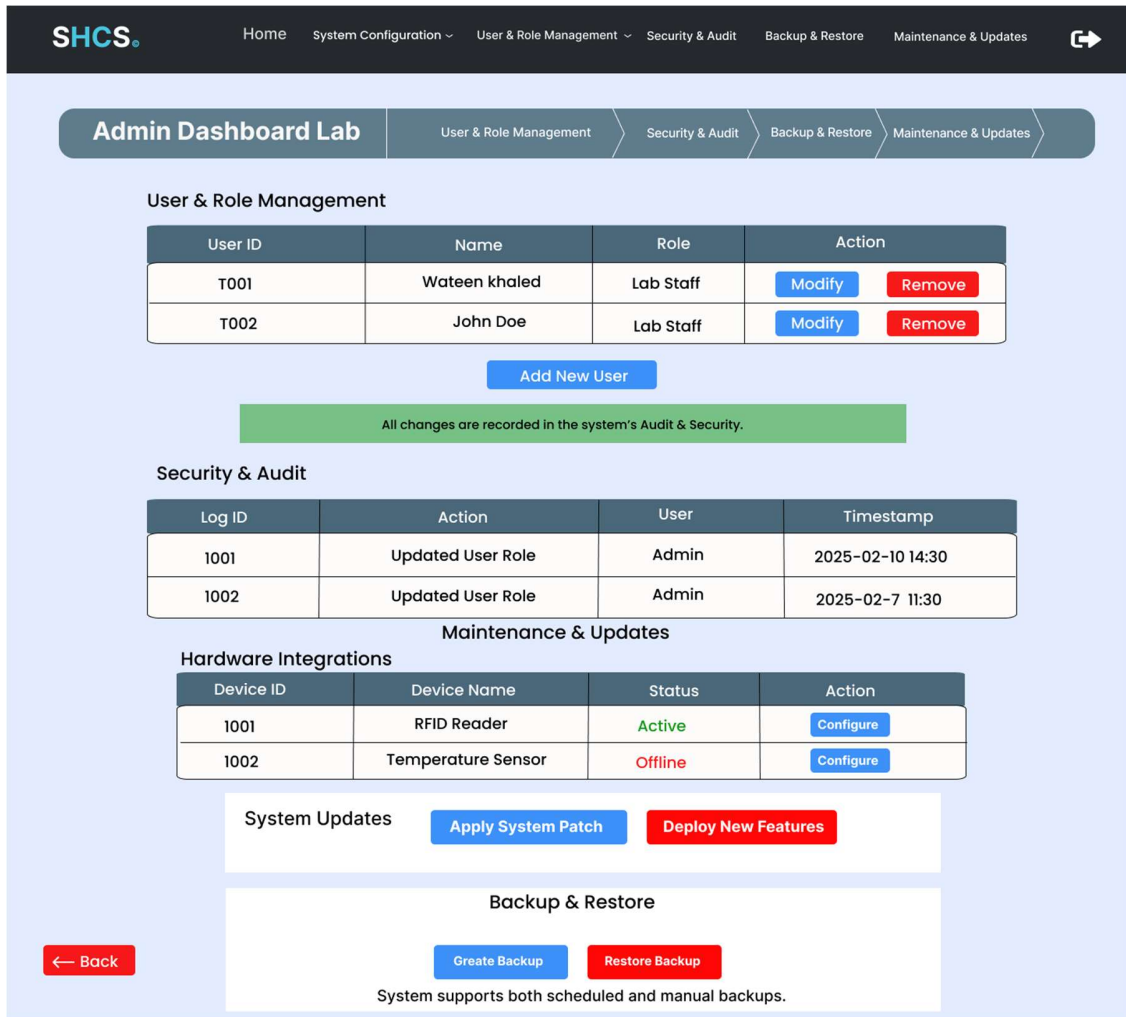


Figure 3.6 : Admin Dashboard Lab Interface

5.7 Summary

This chapter highlights the structure design of the Smart Health Care System ensuring seamless integrations by the system components. Starting with the sequence diagrams for each role. Next the class diagrams for each one of them to make it simple as seeable. Then we have the ERD diagrams to provide an overview of database architecture, and we have the context diagram that shows the departments that our system will cover and specialized with it. Lastly, we have the interface design reflect the smoothly and simplicity of the work flow for the users. Overall, chapter four establishes the structural framework for the Smart Health Care System.

6. Chapter Five: Conclusion and Future Work

6.1 Conclusion

This project came up with a Smart Health Care System that addresses the critical operational concerns by bringing various departments together (Operating Room, Sterilization, Maintenance, and Laboratory) into one system. The system depends first on RFID technology to track trace instruments and equipment in real-time allowing for verification, management of sterilization workflows, scheduling of maintenance, and oversight of lab samples.

Fundamental aspects include:

1. Full visibility: Foreach role they can see where the instruments, equipment or sample's location and state.
2. Automated verifications: Thes system provide an automated identifies for the missing instruments, incomplete trays, required equipment and easily check what the operation needs or require.
3. Keep on track: the system can detect temperature changes in lab storage record it and alert the staff for it, enable the lab staff to move or have better understand for the test result.
4. Risk management: The system ensures everything in their place for the operation before it gets started.

These features are worked together to provide as much help as possible for the hospital staff to increase efficiency, reduce the wasted time and manage the risk as much as possible.

6.2 Future Work

In the future, there are a few ways to grow and improve our Smart Health Care System:

1. Scale the Equipment tracking: The system currently tracks the equipment only in OR and maintenance departments. Future versions will cover the entire hospital departments by adding more RFID readers across all the hospital.
2. Material Management: introduce a material track by an automated way to cut the usage from the storage keeping eye on levels and alerts the staff for restock.

6.3 References

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